Indian Institute of Technology Madras

Overview: Optimization Methods

Content

Introduction to optimization and different forms of optimization

- Unconstrained vs Constrained optimization
- Linear programming (LP)
- Quadratic programming (QP)
- Nonlinear programming (NLP)
- Numerical Optimization Techniques

Optimization

Elements

- Linear function: $f(x)$ is a linear function if it satisfies the following condition:

$$f(\alpha\, x_1 + \beta\, x_2) = \alpha f(x_1) + \beta f(x_2)$$

- Convex function: $f(x)$ is a convex function if

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)x_2, \quad \lambda \in [0,\ 1]$$

- Nonlinear function: Functions that are not linear. Quadratic functions $f(x) = x^T\, Q\, x$ is one type of nonlinear function

Optimization
Elements

- Mathematical Optimization (or Mathematical Programming):
Select the best option or a set of options from the available set
- Consider an optimization problem with decision variables $\mathbf{x} = [x_1, x_2, ..., x_n]^T$

$$\begin{aligned} \max_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{Ax} \leq \mathbf{b} \\ & g(\mathbf{x}) \leq 0 \end{aligned}$$

or

$$\max_{x_1, x_2, ..., x_n} \quad f(x_1, x_2, ..., x_n)$$

- $f(\mathbf{x})$ is called *cost function* or *objective function* or *loss function*
- $\mathbf{Ax} \leq \mathbf{b}$ and $g(\mathbf{x}) \leq 0$ are *constraints*

Optimization
Constraints

- Types of constraints
  - Linear Constraints

$$\mathbf{A}\mathbf{x} \leq \mathbf{b} \quad \text{(Inequality)}$$

$$\mathbf{A}_{eq}\mathbf{x} = \mathbf{b}_{eq} \quad \text{(Equality)}$$

  - Inequality Constraints

$$\mathbf{g}(\mathbf{x}) \leq \mathbf{0} \quad \text{(Inequality)}$$

$$\mathbf{h}(\mathbf{x}) = \mathbf{0} \quad \text{(Equality)}$$

  - Bounds: $\mathbf{x}_{lb} \leq \mathbf{x} \leq \mathbf{x}_{ub}$
- $\mathbf{x} \in \mathcal{S}$ For example, $\mathcal{S} = \{-1, 0, 1\}$

Optimization

Types

- Unconstrained optimization: No constraints on decision variables
- Constrained Optimization: Constrained on decision variables
- Static optimization: No time involved
- Dynamic optimization: Time is involved
- Types of function or variable set smoothness:
  - Continuous Optimization: Decision variables take continuous real values
  - Integer optimization: Decision variables take integer values

Optimization

Types

- Linear Programming (LP)

$$\begin{aligned} \max_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{A}_{eq}\mathbf{x} = \mathbf{b}_{eq} \end{aligned}$$

- Quadratic Programming (QP)

$$\begin{aligned} \max_{\mathbf{x}} \quad & \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} + \mathbf{c}^T\mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{A}_{eq}\mathbf{x} = \mathbf{b}_{eq} \end{aligned}$$

- Nonlinear Programming (NLP) or Nonlinear Optimization

Motivation

Static Optimization

- Static Optimization: An optimal number or finite set of numbers
- Static Optimization:

$$\max_{x} \quad f(x) \tag{1}$$

- Assumptions: $f(x)$ is continuously differentiable
- First order necessary condition: $\frac{\partial f}{\partial x}(x^*) = 0$
- Second order necessary condition: $\frac{\partial^2 f}{\partial x^2} \leq 0$
- Example: The operating point x^* that maximizes the profit $f(x)$, where x: $\#$ of units

Motivation

Static Optimization

- Example

$$\max_{x} \quad 1000000 + 4000x - x^2 \tag{2}$$

- $\frac{\partial f}{\partial x}(x^*) = 0 \Rightarrow 4000 - 2x = 0$
  $x^* = 2000$
- $\frac{\partial^2 f}{\partial x^2} = -2 < 0$

Motivation

Static Optimization

- Static Optimization with several variables

$$\max_{x_1,\ldots,x_n} \quad f(x_1,\ldots x_n) \tag{3}$$

- Example: A plant can produce n items. Find the operating point $x_1,\ldots x_n$ that maximizes the profit $f(x_1,\ldots x_n)$

Motivation

Static Optimization

- Example

$$\max_{x} \quad 1000000 + 300x_1 + 500x_2 - x_1^2 - x_2^2 \tag{4}$$

- $\frac{\partial f}{\partial x}(x_1^*) = 0 \Rightarrow 300 - 2x_1 = 0 \Rightarrow x_1^* = 150$
- $\frac{\partial f}{\partial x}(x_2^*) = 0 \Rightarrow 500 - 2x_2 = 0 \Rightarrow x_2^* = 250$
- $\frac{\partial^2 f}{\partial x_1^2} = \frac{\partial^2 f}{\partial x_2^2} = -2 < 0$
- These conditions are correct??
- Karush-Kuhn-Tucker Conditions (KKT COnditions)

Constrained Optimization

Equality Constraints and Lagrange Function

- Consider the following optimization problem

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0,\ i = 1, 2, \ldots, m \end{aligned} \tag{5}$$

- Constraint optimization to Unconstrained optimization
- Lagrange function with Lagrange multipliers $\lambda_1, \ldots, \lambda_m$

$$l(x, \lambda) = f(x) + \sum_{i=1}^{m} \lambda_i h_i(x)$$

Constrained Optimization

Karush–Kuhn–Tucker (KKT) Conditions

- First-order necessary condition

$$\nabla f(x^*) + \lambda_1^* \nabla h_1(x^*) + \ldots + \lambda_m^* \nabla h_m(x^*) = 0$$

$$h_i(x^*) = 0, \; i = 1, 2, \ldots, m$$

- Second-order necessary condition

$w^T l_{xx}(x^*, \lambda^*) w \geq 0, \forall w$ such that $\nabla h_i(x^*).w = 0, \; i = 1, 2, \ldots, m$

where

$$l_{xx}(x^*, \lambda^*) = \nabla^2 f(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla^2 h_i(x^*)$$

$l_{xx}(x^*, \lambda^*)$ is positive definite on the tangent space defined by $\nabla h_i(x^*).w = 0$ the tangent space

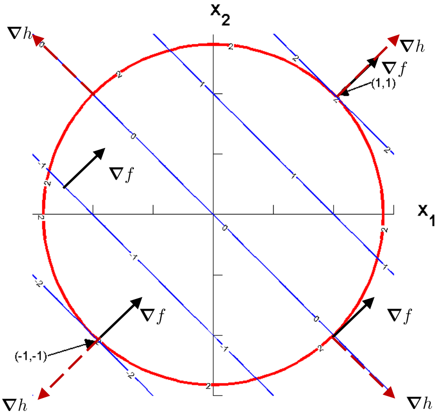
Example: Constrained Optimization

- Problem:

$$\min \quad x_1 + x_2$$

$$\text{s.t.} \quad x_1^2 + x_2^2 - 2 = 0$$

- $\nabla f = [1, 1]^T \ \ \nabla h = [2x_1, 2x_2]^T$



Constrained Optimization: Example

KKT Conditions

- Constrained optimization problem:

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0,\ i = 1, \ldots, m \\ \text{s.t.} \quad & g_j(x) \leq 0,\ j = 1, \ldots, n \end{aligned}$$

  where f, h_i, and g_j: smooth, and real-valued functions on a subset of $\mathbb{R}^n$
- x is a feasible point, if it satisfies the equality and inequality constraints
- A constraint j is said to be active if $g_j(x) = 0$ at any point x
- $A(x)$: A set of all active constraints at any point x
- For a feasible point x and the active set $A(x)$, if the gradients ∇h_i, and $\nabla g_j, \forall j \in A(x)$ are linearly independent,they satisfy the Linear Independence Constraint Qualification (LICQ).

Constrained Optimization

KKT Conditions

- Lagrangian for the problem with the Lagrange multipliers λ and μ:

$$l(x, \lambda, \mu) = f(x) + \sum_{i=1}^{m} \lambda_i h_i(x) + \sum_{j=1}^{n} \mu_j g_j(x)$$

- First order Necessary condtions (Karush-Kuhn-Tucker conditions): For x^* a local solution

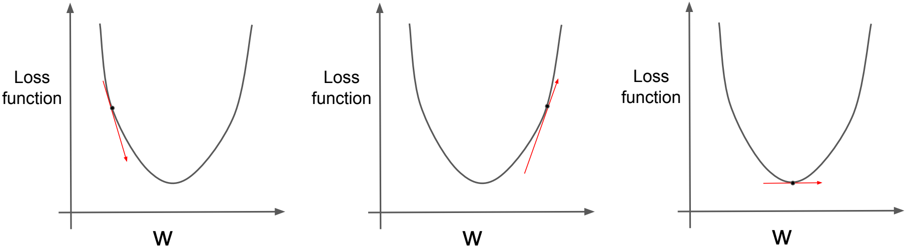
$$\begin{aligned} \nabla_x L(x^*, \lambda^*, \mu^*) &= 0 \\ h_i(x^*) &= 0, i = 1, \ldots, m \\ g_j(x^*) &\leq 0, i = 1, \ldots, n \\ \lambda_i &> 0,\ \mu_j \geq 0 \\ \mu_j(x^*) g_j(x^*) &= 0 \text{ (Complementarity condition)} \end{aligned}$$

- Complementarity condition: Ensures $\mu_j = 0, \forall j \notin A(x)$

Optimization in Machine Learning and Deep Learning

- Central Idea to DS/ML/AI: Minimize or maximize an objective function
- Deep learning objective: minimize the loss function or objective function
- Direct solution & Iterative solution
- Arthur Samuel's paradigm:
  "Mechanism" to improve performance by tweaking weights (parameters)
- Loss function can be tweaked by varying the model parameters
  Million dimensional space!

Gradient Descent



$\nabla Loss w < 0$	$\nabla Loss w > 0$	$\nabla Loss w = 0$
Increment w	Decrement w	Optimum w
$w = w + \Delta w$	$w = w - \Delta w$	$\Delta w = 0$ (stationary point)

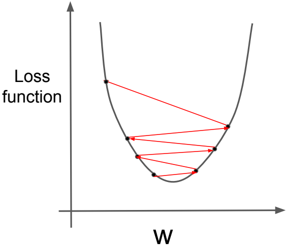
What should be $\Delta\, w$?

$\Delta w = -\eta \nabla Loss w \Rightarrow\ w - \eta \nabla Loss w$, where η : Learning rate

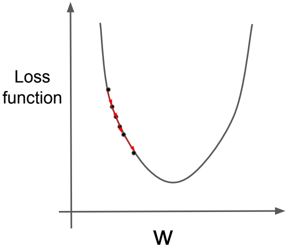
Gradient Descent

Learning Rate

Learning rate is too big⇒ Oscillation diverges



Learning rate is small⇒ Long time to convergence



- Learning rate should be:
  - Near local solution: proceed quickly with small learning rate
  - Far from local solution: proceed quickly with large learning rate

Gradient Descent

Three types

Batch GD

- Use the entire data set for computing the gradient
- Update rule

$$w = w - \eta \nabla_w \mathrm{L}(\mathrm{w})$$

- Slow
- large data set does not fit in memory

Stochastic GD (online)

- Use data point (x_i, y_i) for computing gradient
- Update rule

$$w = w - \eta \nabla_w \mathrm{L}\,(w, x_i, y_i)$$

- Much faster
- Update with high variance L fluctuates heavily

Mini-batch GD

- Use a subset of data points for computing the gradient
- Update rule $w = w - \eta \nabla_w \mathrm{L}(w, x_{i:i+n}, y_{i:i+n})$
- Best of both methods
- Reduces variances in parameter updates

Algorithm of choice: Mini-batch gradient descent for NN and DL

Mini-batch GD is often referred to as SGD

Typical batch sizes of 50 or 256 are used

Gradient Descent

Momentum SGD

- SGD: Sometimes slow
- How do we accelerate the learning rate?
- Momentum approaches:
  - Use concept of momentum from physics
  - v: velocity variable
- Update rule

$$w_{k+1} = w_k - \eta \nabla_w \mathrm{L}(w, x_i, y_i) + \beta v$$

- v: accumulates the past gradients
- $\beta \in [0, 1)$: momentum parameter
- Larger β, current direction is effected by the more past gradients

Gradient Descent

Stochastic GD: Summary

- The element for applying stochastic GD
  - The loss function form
  - A way to compute gradients wrt parameters
  - Initialization for parameters and learning rate and other hyperparameters

[2] Schaul, et al. "No more pesky learning rates." International Conference on Machine Learning. 2013.

Indian Institute of Technology Madras

Probability and Statistics for Data Science and AI

Nirav Bhatt
Email: niravbhatt@iitm.ac.in

Module: Probability and Statistics for DSAI

- Probability for DSAI
  - Revisiting: High school probability topics
  - Conditional Probability
  - Random Variables, Expectation, Distributions, and Random Processes
  - Applications of concepts in Probability for DSAI
- Statistics for DSAI
  - Descriptive statistics and Visualization
  - Inferential Statistics: Frequentist inference, Bayesian inference, Hypothesis testing
  - Applications of concepts in Statistics for DSAI

Outline

- Experiments and Random Phenomena
- Revisit to high school probability topics
- Random variables
- Important probability distributions and properties
- Descriptive statistics

Statements we make!

- Mom, I will be there in 10 minutes.
- Today, it is very humid, it will definitely rain.
- Amazon parcel will most likely to deliver today.
- I may make it to your party tomorrow.
- Machchi, You are less likely to get Corona.
- It is impossible to get 100% in this exam

Statements we make!

- How do we make these statements?
  - Past experience?
  - Some information available to you?
- Collected data all the time about different phenomena around us
- Do reasoning subconsciously
- Make these statements

Experiments

- **Experiment I:** Tossing a coin
  - Fair coin, Fair and same tossing machine, constant wind speed, temperature or any other variables that can affect the outcome of tossing a coin
  - Outcome at each tossing:

  - Observations

Experiments

- **Experiment II:** Measuring concentration of 1 M glucose in bottle by a set of students
  - Same day, same equipment, constant temperature or any other variables that can affect the outcome
  - Outcomes: 0.98, 1, 1.2, 0.85, 0. 97, 1.02, 1.1, 0.95, 0.89, 1.06, ...
  - Observations

Experiments

- **Experiment III**
  - Determine (integer) age of a particular student in the class room
    Using date of birth on ID card
  - Outcomes: 18, 18, 18, 18, 18, 18, 18, 18, 18
  - Observations

Experiments and Variability

- Experiments I and II
  - Same experimental conditions
  - Different outcomes
  - Outcome cannot be predicted with certainty
  - Encountered variability
- Experiment III:
  - Same experimental conditions
  - Identical outcomes
  - Outcome can be predicted with certainty
  - Deterministic in nature

Variability and Random Phenomena

- Variability
  - Something that cannot be controlled

  - “Identical" conditions: Different outcomes of the phenomenon

  - Do not product exactly the same outcome

- Experiments I and II
  - Each toss either Head or Tail
  - Student measured different glucose concentrations for 1 M bottle
- Random Phenomenon:
  A phenomenon in which “randomness” is involved and outcomes cannot be predicted or controlled

Random Phenomena

- Characterising random phenomena
  - Different outcomes for same experimental conditions
  - Variability associated with different outcomes

    Is it possible to characterize?
  - Example I : Number of students attending BT5450 class on Friday?

    Finite number of outcomes: (0,. . . ., total number of students taking the course) — Discrete
  - Example II: Average height of students attending BT5450 class on Friday in cm?

    Infinite number of outcomes: Any number in an interval — Continuous

Random Phenomenon

- Deterministic Phenomenon
  - A phenomenon whose outcome can be predicted with a very high degree of confidence for the same experimental conditions
  - Example: Age of a person in year based on birth date in driving licence
- Random Phenomenon
  - A phenomenon which can have several possible outcomes for the same experimental conditions
  - Outcome can be predicted with limited confidence
  - Example: Arrival time of 9.30 am bus at bus-stand in India

Characterizing random phenomena

- Sources of error in observed outcomes
  - Model error: Lack of knowledge of generating process

    e.g. Development of unstructured kinetic models
  - Measurement error: Errors in sensors used for observing outcomes

    e.g. Measuring weight on different weighting machines
- Types of random phenomena
  - Discrete: Outcomes are finite
  - Continuous: Infinite number of outcomes

Random Phenomena

- Our interest: Variability associated with model and measurement errors
- Need a framework allows to understand and quantitatively characterize random phenomena

Random Experiment

- An experiment that can result in different outcomes when it is repeated under identical conditions every time
- Analysis of such a system: need to understand set of all possible outcomes

Sample space and events

- Sample space: The set of all possible outcomes of a random experiment

- Event: Subset of the sample space is an event

Concept of Counting

Permutation

- Example: A,C,G,T. How many ways they can be arranged without repeating them?
  ACGT,ACTG,ATCG,ATGC AGCT,AGTC
  CAGT,CATG,CGAT,CGTA,CTGA,CTAG
  TAGC,TACG,TCAG,TCGA,TGAC,TGCA
  GCAT,GCTA,GATC,GACT,GTAC,GTCA
- Permutations : Arrangement of items in an orderly manner
- n=4 (A,C,G,T), $4!=4 \times 3 \times 2\times1=24$
- **Distinct *n* objects can be arranged in n ! (factorial) ways**

Concept of Counting

Permutation

- Example: How many 4 digits number can be created from 0,1,...,9 without repetition of digits?
- **Permutations (without repetition): From distinct n objects choose r:** $P_r^n = \dfrac{n!}{(n-r)!}$
- $P_4^{10} = \dfrac{10!}{(10-4)!} = \dfrac{10!}{6!} = 10 \times 9 \times 8 \times 7 = 5040$
- What about permutation with repetition?
- $10 \times 10 \times 10 \times 10 = 10^4 = 10000$
- **Permutations (with repetition): From distinct n objects choose r,** n^r

Concept of Counting

Permutation

- How many the same length arrangements of “Machchi" are possible?

Concept of Counting

Permutation

- **Permutations with multiple items:** n **objects with items of** k_1 **type,** k_2 **type,**$\ldots,$ k_k**type**

$$\frac{n\,!}{k_1\,!k_2\,!\ldots k_k\,!}$$

- How can a family 5 can be arranged on circular table?
- **Permutations (Clockwise and anti-clockwise are different)**
- **Permutations (Clockwise and anti-clockwise are same)**

Combinations

order doesn't matter

- Choosing a hockey team from 17 players?

Combinations

order doesn't matter

- Combination (without repetitions): the number of combinations in which r items can be chosen from a set of n items $C_r^n = \frac{n!}{r!(n-r)!} = \frac{P_r^n}{r!}$
- six Gelato (Sorbet) flavors in the shop, the number of combination in which three scoops of Gelato (Sorbet) can be chosen?

$$C_3^6 = \frac{(6+3-1)!}{3!(6-1)!} = 56$$

- Combinations with repetition (n items, choosing r items)

$$C_r^n = \frac{(n+r-1)!}{r!(n-1)!}$$

Exclusive and Independent Events

- Mutually exclusive events: Two events are mutually exclusive if occurrence of one precludes occurrence of the other
- Independent events: Two events are independent if occurrence of one has no influence on occurrence of other

Different types of events

- Consider two events E and F of a sample space $\boldsymbol{\Omega}$

New Events

Union ($E \cup F$): A set consisting all elements in either E or F (or both)
Intersection ($E \cap F$): A set consisting all elements in both E and F
Null event (Mutually exclusive): $E \cap F = \{\} \rightarrow$ Events E and F cannot both occur

Different types of events

- Consider two events E and F of a sample space $\boldsymbol{\Omega}$

New Events

Complement event (E^c): E^c is the set of all the elements in $\boldsymbol{\Omega}$ but are not in E, i.e. $E \cap E^c = \Phi$
Difference event ($E - F$): The set of all the elements in $\boldsymbol{\Omega}$ that are in E but are not in F, i.e. $E \cap F^c$

Introduction to Probability

Algebra of Events

- Consider two events E and F of a sample space Ω

Rules

Commutative law: $E \cup F = F \cup E$, $E \cap F = F \cap E$
Associative law: $(E \cup F) \cup G = E \cup (F \cup G)$, $(E \cap F) \cap G = E \cap (F \cap G)$
Distribution law: $(E \cup F) \cap G = (E \cap G) \cup (F \cap G)$
$(E \cap F) \cup G = (E \cup G) \cap (F \cup G)$

Probability Measure

- Probability measure is a function that assigns a real value to every outcome of a random phenomena which satisfies following axioms
  - $P(S) = 1$ (one of the outcomes should occur)
  - $0 \leq P(A) \leq 1$ (Probabilities are non-negative and less than 1 for any event A)
  - For two mutually exclusive events A and B

$$P(A \cup B) = P(A) + P(B)$$

Conditional Probability

Conditional Probability

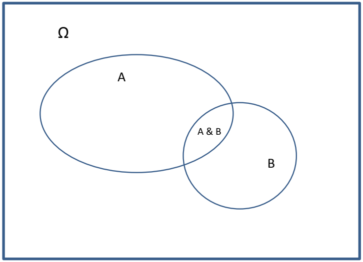
- If two events A and B are not independent, then information available about the outcome of event A can influence the predictability of event B

Definition

The probability of the occurrence of an event subject to the hypothesis that another event(s) has occurred

- $P(E|F)$: the event E will occur given that the event F has occurred.
- Assumption: The event F is not impossible event.
- Computation of $P(E|F)$:

$$P(E|F) = \frac{P(E \cap F)}{P(F)}, \; P(F) > 0$$

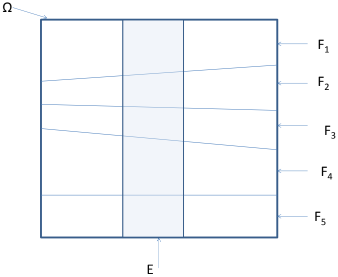


- Example: two (fair) coin toss experiment
  - Event A : First toss is head = {HT, HH}
  - Event B : Two successive heads ={HH}
  - P(B)=0.25 (no information)
  - Given event A has occurred, P(B/A)=0.5

Bayes' formula

- Direct computing the probability of an event is difficult. How?
- Consider mutually exclusive events F_1, F_2,...,F_n of $\mathbf{\Omega}$ such that

$$\bigcup_{i=1}^{n} F_i = \mathbf{\Omega}$$



- For an event E,

$$E = \bigcup_{i=1}^{n} (E \cap F_i)$$

Bayes' formula contd.

- The probability of the event E:

$$
\begin{aligned}
P(E) &= \sum_{i=1}^{n} P(E \cap F_i) \\
&= \sum_{i=1}^{n} P(E|F_i)P(F_i)
\end{aligned}
$$

- $P(F_i)$ given the event E has occurred?
- Bayes' formula:

$$
P(F_i|E) = \frac{P(E \cap F_i)}{P(E)} = \frac{P(E|F_i)P(F_i)}{\sum_{i=1}^{n} P(E|F_i)P(F_i)}
$$

-

Independent Events

- Two events E and F are said to be *independent* if

$$P(E \cap F) = P(E)P(F)$$

- If they are not independent, they are said to be dependent
- Some results:
  - If E and F are independent, then so are E and F^c.
  - The events F_1, F_2,...,F_n are independent if for every subset $F_{1'}$, $F_{2'}$,...,$F_{r'}$, $r' \leq n$, of these events

$$P(F_{1'} \cap F_{2'} \cap \ldots F_{r'}) = P(F_{1'})P(F_{2'}) \ldots P(F_{r'})$$

Random Variables

- Experiment 1: The testing of three components for quality (*N*: Non-defective, and *D*: Defective)
  $\mathbf{\Omega} = \{NNN, NND, NDN, DNN, NDD, DND, DDN, DDD\}$
- Each point in $\mathbf{\Omega}$ assigned a numerical value (0,1,2,3)
  $X(NNN)$ =0; $X(NND)$=1,
  $X(NDN) = 1$,$X(DNN)$=1;
  $X(NDD) = 2$, $X(DND) = 2$,
  $X(DDN) = 2$, $X(DDD)$=3

Random Variables

- Experiment 1: The testing of three components for quality (*N*: Non-defective, and *D*: Defective) $\mathbf{\Omega} = \{NNN, NND, NDN, DNN, NDD, DND, DDN, DDD\}$
- Each point in $\mathbf{\Omega}$ assigned a numerical value (0,1,2,3) $X(NNN)$ =0; $X(NND)$=1, $X(NDN) = 1$,$X(DNN)$=1; $X(NDD) = 2$, $X(DND) = 2$, $X(DDN) = 2$, $X(DDD)$=3
- Experiment 2: Choosing three fruits from the basket having apples (*A*) and oranges (*O*) $\mathbf{\Omega} = \{AAA, AAO, AOA, OAA, AOO, OAO, OOA, OOO\}$
- Each point in $\mathbf{\Omega}$ assigned a numerical value (0,1,2,3) $X(AAA)$ =0; $X(AAO)$=1, $X(AOA) = 1$,$X(OAA)$=1; $X(AOO) = 2$, $X(OAO) = 2$, $X(OOA) = 2$, $x(OOO)$=3

Random Variables

- Experiment 1: The testing of three components for quality (N: Non-defective, and D: Defective)
  $\mathbf{\Omega} = \{NNN, NND, NDN, DNN, NDD, DND, DDN, DDD\}$
- Each point in $\mathbf{\Omega}$ assigned a numerical value (0,1,2,3)
  $X(NNN)$ =0; $X(NND)$=1, $X(NDN) = 1$,$X(DNN)$=1; $X(NDD) = 2$, $X(DND) = 2$, $X(DDN) = 2$, $X(DDD)$=3

- Experiment 2: Choosing three fruits from the basket having apples (A) and oranges (O)
  $\mathbf{\Omega} = \{AAA, AAO, AOA, OAA, AOO, OAO, OOA, OOO\}$
- Each point in $\mathbf{\Omega}$ assigned a numerical value (0,1,2,3)
  $X(AAA)$ =0; $X(AAO)$=1, $X(AOA) = 1$,$X(OAA)$=1; $X(AOO) = 2$, $X(OAO) = 2$, $X(OOA) = 2$, $x(OOO)$=3

Definition

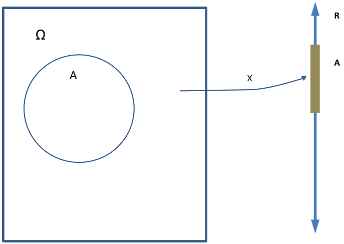
A function that associates a unique real number corresponding to every outcome of sample space.

Random Variables

- Formal definition:

Random Variable

A function $X : \mathbf{\Omega} \rightarrow \mathbf{R}$ is a random variable if and only if $P(\{\omega \in \mathbf{\Omega} : X(\omega) \leq y\})$ exists for all choices of $y \in \mathbf{R}$.

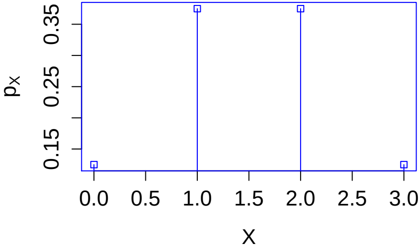


- Coin toss sample space [H T] mapped to [0 1].
- If the sample space outcomes are real valued no need for this mapping (eg. throw of a dice)
- Allows numerical computations such as finding expected value of a RV
- Discrete RV (throw of a dice or coin) Continuous RV (sensor readings, time interval between failures)
- Associated with the RV is also a probability measure

Random Variables

- Example: Probability measure

$$P_X(X=0)=\frac{1}{8},$$
$$P_X(X=1)=\frac{3}{8},$$
$$P_X(X=2)=\frac{3}{8},$$
$$P_X(X=3)=\frac{1}{8}$$



- Some observations: X is a finite integer number, and discrete
- Discrete random variable: X *whose set of possible values forms a discrete set*
- Set of possible values : a finite or infinite sequence
- P_X: Probability mass function (PMF): $P_X(X=x_i)=p_i$
- PMF also follows properties of a general probability

Random Variables

- Examples:
  - X: Length of a randomly selected telephone call
  - X: Life of a randomly selected bicycle in IIT
  - X: Height, weight of a randomly selected students etc
- Observation: X: continuous in nature, and can take on any value in some interval (a, b)

Random Variables

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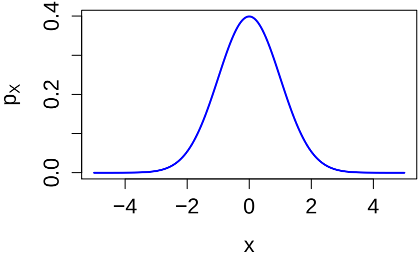
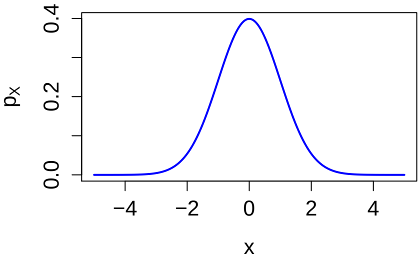


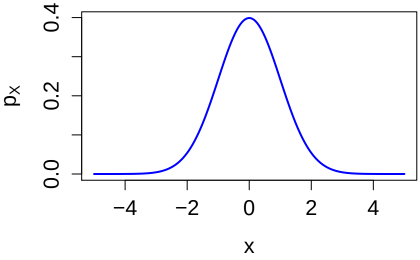
Random Variables



- PMF will not provide any useful information
- X continuous variable if there is a function $f(x) \geq 0$ so that any interval $-\infty \leq a \leq b \leq \infty$,

$$P_X(a \leq X \leq b) = \int_a^b f(x)dx$$

Random Variables



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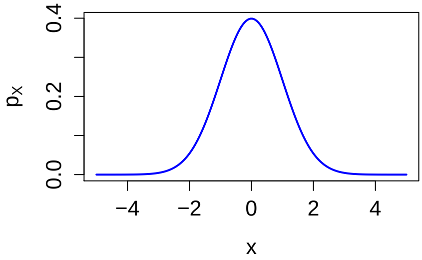
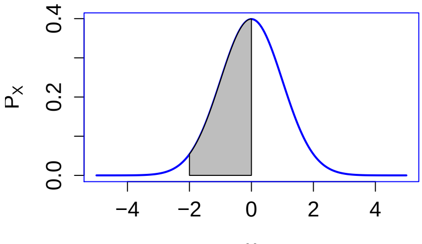
- $P(-2 \leq X \leq 0)$?

Random Variables





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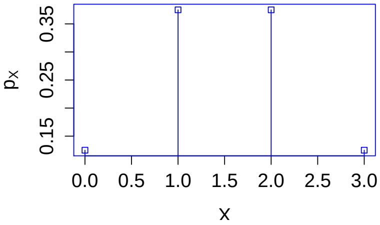
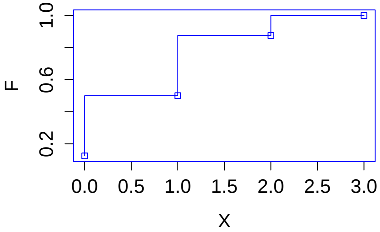
- $P(-2 \leq X \leq 0)$= Area under the curve

Random Variables





- Cumulative distribution function, c.d.f. of X for any real number a

$$F(a) = P(X \leq a) = \sum_{all\ x \leq a} p(x)$$

- Example: F(1) =0.5, F(2)=7/8
- c.d.f. for continuous X

$$\begin{aligned} F(a) &= P(X \leq a) \\ &= P(X \in (-\infty, a])) \\ &= \int_{-\infty}^{a} f(x)dx \end{aligned}$$

- Relationship between c.d.f and p.d.f

$$\frac{dF(a)}{da} = f(a)$$

Random Variables

- Game of rolling a balanced die:
  - Result: 2, 3 or 4; *Win: Rs. 10*
  - Result: 5; *Win: Rs. 20*
  - Result: 1, 6; *Win: Rs. -30*
- Should you play the game?
- $\mathbf{\Omega} = \{1, 2, 3, 4, 5, 6\}$
- X: the payoff amount in the game
  $X(1) = X(6) = -30$;
  $X(2) = X(3) = X(4) = 10$; $X(5) = 20$
- Probability mass function
  - $P_X(X = -30) = 1/3$
  - $P_X(X = 20) = 1/6$
  - $P_X(X = 10) = 1/2$

- Expected the payoff amount after the n games

$$E(Payoff) = (20\,\frac{1}{6} + 10\,\frac{1}{3} - 30\,\frac{1}{3})\,n$$

$$E(Payoff) = -\frac{10\,n}{6}$$

Random Variables

- Expectation or expected value of X ($E[X]$): the average value of X in a large number of trails
- Not a value X could possible take
- For discrete random variables:

$$E[X] = \sum_i x_i P(X = x_i) \tag{1}$$

- For continuous random variables:

$$E[X] = \int_{-\infty}^{\infty} x\, f(x)\, dx \tag{2}$$

Random Variables

- $E[X]$: The weighted average of the possible values of X, often denoted as μ

  No information about the spread of these values

- If we quantify the spread using a function $\phi(x)$

$$\phi(x) = (x - \mu)^2 \tag{3}$$

Definition: Variance

If X is a random variable with mean μ, the variance of X, denoted by $Var(X)$, is defined by

$$\begin{aligned} Var(X) = E[(X-\mu)^2] &= \sum_{x \in X(S)} (x-\mu)^2 f(x) \\ &= E[X^2] - (E[X])^2 \end{aligned}$$

- The standard deviation of X is the square root of its variance

Random Variables

- Some properties of variance
  - For constant c, $Var(X + c) = Var(X)$
  - $Var(cX) = c^2 Var(X)$
  - $Var(X_1 + X_2) = Var(X_1) + Var(X_2)$
- Variance for a continuous random variable

$$Var(X) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx \tag{4}$$

Random Variables

- Some properties of variance
  - For constant c, $Var(X + c) = Var(X)$
  - $Var(cX) = c^2 Var(X)$
  - $Var(X_1 + X_2) = Var(X_1) + Var(X_2)$

Random Variables

- For two random variables X and Y

Definition: Covariance

If $E[X] = \mu_X$ and $E[Y] = \mu_Y$, the covariance of X and Y,

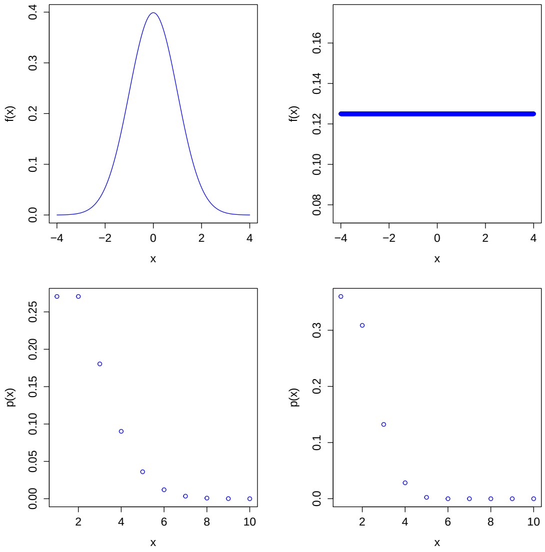
$$\begin{aligned} Cov(X,Y) &= E[(X-\mu_X)(Y-\mu_Y)] \\ &= E[XY] - E[X]E[Y] \end{aligned}$$

- Properties:

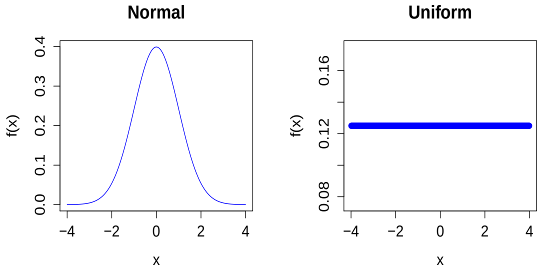
$$\begin{aligned} Cov(X+Z,Y) &= Cov(X,Y) + Cov(Z,Y) \\ Cov(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j) &= \sum_{i=1}^{n}\sum_{j=1}^{m} Cov(X_i,Y_j) \\ Var(\sum_{i=1}^{n} X_i) &= \sum_{i=1}^{n} Var(X_i) + \sum_{i=1}^{n}\sum_{j=1,j\neq i}^{m} Cov(X_i,Y_j) \end{aligned}$$

- X and Y are independent, then $Cov(X,Y) = 0$

Random Variables

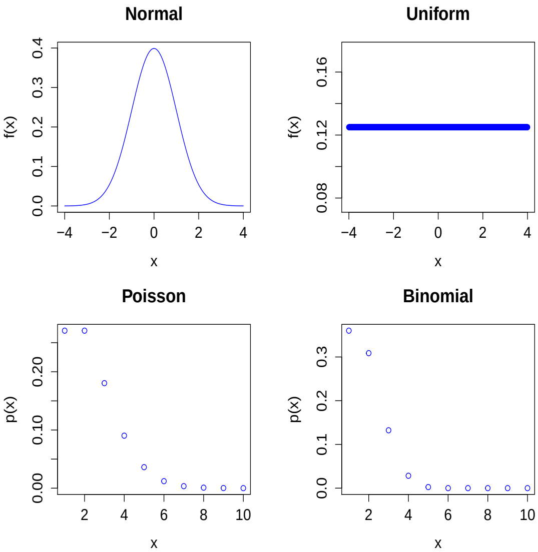


Random Variables



$\leftarrow$ Continuous distributions

Random Variables



← Continuous distributions

← Discrete distributions

Random Variables

- Distributions:
  - Discrete: Poisson, Binomial, Geometric, hyper-geometric etc.
  - Continuous: Normal and its variants,lognormal Exponential, Beta etc.

Random Variables

- Distributions:
  - Discrete: Poisson, Binomial, Geometric, hyper-geometric etc.
  - Continuous: Normal and its variants,lognormal Exponential, Beta etc.

Binomial Distribution

- Example:
  - Every minute 24 births in India, X be the number of female births=12
  - In pass and fail course, X is either pass or failure
- Experiment's outcome: Either Success or failure.
- If $X = 1$ for success, and $X = 0$ for failure,

$$P(X = 0) = 1 - p$$
$$P(X = 1) = p$$

- For n independent trails, X represents $\#$ of successes in the n trails, $X \sim b(n, p)$, $b(n, p)$ is Binomial distribution with parameters n and p.
- PMF

$$P(X = i) = \binom{n}{i} p^i \, (1 - p)^{n-i}, \; i = 0, 1, \ldots, n$$

Binomial Distribution

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$$P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i},\ i = 0, 1, \ldots, n$$

- Mean and variance for X

$$\begin{aligned} E[X] &= np \\ Var[X] &= np(1 - p) \end{aligned}$$

Poisson DistributionRandom Variables

- If X is a Poisson random variable with parameter $\lambda > 0$, PMF of X
$$P(X = i) = \frac{e^{-\lambda}\lambda^i}{i!},\ i = 0, 1, 2, \ldots, n$$
- Mean and variance of X

$$\begin{aligned} E[X] &= \lambda \\ Var[X] &= \lambda \end{aligned}$$

NormalRandom Variables Distribution

- Consider a density function $f(x)$, $-\infty < x < \infty$:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{5}$$

  with parameters μ and σ.
- $X \sim \mathbb{N}(\mu, \sigma^2)$ if $f(x)$ is the density of X
- Normal distribution is symmetric about μ
- Examples
  - The height, weight of students in IIT Madras
  - The velocity of molecule in the gas
- Some results:
  - $E[Y] = E[a + b\,X] = a + bE[X] = a + b\,\mu$
  - $Var(Y) = Var(a + bX) = b^2\,Var(X) = b^2\,\sigma^2$

Normal Distribution

- Properties of normal desntiy function
  - $f(x) > 0$, and a nonempty set (a, b)

$$P(X \in (a, b) = Area_{(a,b)} f(x) \tag{6}$$

  - $f(x)$ is symmetric about μ, i.e., $f(\mu + x) = f(\mu - x)$
  - $f(x)$ decreases as $|x - \mu|$ increases
  - For all choice of *mu* and σ

$$\begin{aligned} P(\mu - \sigma < X < \mu + \sigma) &= 0.683 \\ P(\mu - 2\,\sigma < X < \mu + 2\,\sigma) &= 0.954 \\ P(\mu - 3\,\sigma < X < \mu + 3\,\sigma) &= 0.997 \end{aligned}$$

i.e., Given a probability distribution function for any one normal distribution, One can compute probability for any RV with μ and σ

Normal Distribution

- Define a random variable Z:

$$Z = \frac{X - \mu}{\sigma} \tag{7}$$

- $X \sim \mathbb{N}(\mu, \sigma^2)$, Z - distribution?
- $Z \sim \mathbb{N}(0, 1)$ (Standard normal distribution)
- Distribution function

$$\mathbf{\Phi}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{z^2}{2}} dz \tag{8}$$

- Some results:

$$\begin{aligned} P(X < b) &= P\left(\frac{X - \mu}{\sigma} < \frac{X - \mu}{\sigma}\right) = \mathbf{\Phi}\left(\frac{b - \mu}{\sigma}\right) \\ P(a < X < b) &= \mathbf{\Phi}\left(\frac{b - \mu}{\sigma}\right) - \mathbf{\Phi}\left(\frac{a - \mu}{\sigma}\right) \end{aligned} \tag{9}$$

Normal Distribution: Chi-square distribution

- $Z_1, Z_2, \ldots, Z_k$ are independent standard normal random variables
- Define a random variable P:

$$P = \sum_{i=1}^{k} Z_i^2 \tag{10}$$

- $P \sim \chi^2(k)$, χ^2 - distribution?
- Useful in hypothesis testing and other testing

Distribution of Sample statistics

- $X_1, X_2, \ldots, X_n$ random variables from $\mathcal{N}(\mu, \sigma^2)$
- Average of these RVs (mean) $\bar{X} = \frac{X_1+X_2+\ldots+X_n}{n}$
- Question: Distribution of $\bar{X}$ (sample stistics) ?

$$E[\bar{X}] = \mu$$
$$Var[\bar{X}] = \frac{\sigma^2}{n}$$

- $\bar{X}\ \mathcal{N}(\mu, \sigma^2/n)$
- $\sigma/\sqrt{n}$: Standard error in $E[\bar{X}]$ estimate
- As $n \to \infty$, $\bar{X} \to \mu$

Normal Distribution: Chi-square distribution

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- $\sigma/\sqrt{n}$: Standard error in $E[\bar{X}]$ estimate
- As $n \to \infty$, $\bar{X} \to \mu$

Central Limit Theorem

- Interpretation: The sum of a large number of independent RV has a distribution that is approximately normal
- it captures the fact that the empirical frequencies of so many natural populations follows a normal curve
- $X_1, X_2, \ldots, X_n$ random variables , independently and identically distributed (i.i.d.) with finite μ and σ.
- Then for large n ($n \to \infty$), the distribution of $X_1 + X_2 + \ldots + X_n \to \mathcal{N}(n\mu, n\sigma^2)$
- The central limit theorem that

$$\frac{X_1 + X_2 + \ldots + X_n - n\,\mu}{\sigma\sqrt{n}}$$

is approximately a standard normal RV; thus, for n large,

$$P\left(\frac{X_1 + X_2 + \ldots + X_n - n\,\mu}{\sigma\sqrt{n}}\right) \approxeq P(Z < x)$$

where Z is a standard normal random variable.

Sample Statistics

- Central tendencies: , Sample median and Sample mode
- Spread (or variability)of data: Sample variance and sample standard deviation

Sample Statistics

- Central tendencies: Sample mean, Sample median and Sample mode
- Spread (or variability)of data: Sample variance and sample standard deviation

Sample Statistics

- Data set (n observation, numerical values): $x_1, x_2, \ldots, x_n$

Sample mean

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n}$$

- Data set(n observation, numerical values): $x_1, x_2, \ldots, x_k$ corresponding frequencies $f_1, f_2, \ldots, f_k$

Sample mean

$$\bar{x} = \sum_{i=1}^{k} \frac{f_i\, x_i}{n}$$

Sample Statistics

- Salary data of a SME company

Employ	Salary/month (Rs.)
1	20,000
2	15,000
3	12,000
4	35,000
5	8,000
6	11,000
7	20,000
8	25,000
9	1,20,000
10	3,15,000

Sample Statistics

- Salary data of a SME company

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- Mean Salary: Rs. 53909.09

Sample Statistics

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- Mean Salary: Rs. 53909.09
- Mean Salary: Rs. 17555.56 (after removing Employ No. 9 and 10)

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6	11,000
7	20,000
8	25,000
9	1,20,000
10	3,15,000
11	12,000

Sample Statistics

- Salary data of a SME company

Employ	Ascending Order	Salary/month (Rs.)
5	1	8,000
6	2	11,000
3	3	12,000
11	4	12,000
2	5	15,000
1	6	20,000
7	7	20,000
8	8	25,000
4	8	35,000
9	10	1,20,000
10	11	3,15,000

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10	11	3,15,000

- Middle Salary: Rs. 20,000
- Most Common salary?

Sample Statistics

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- Most Common salary?
- Common Salaries: Rs. 12,000; Rs. 20,000

Sample Statistics, Cont.

- Data set (n ordered observation from smallest to largest, numerical values): $x_1, x_2, \ldots, x_n$

Sample median

If n: odd, $x_{md} = (n+1)/2$ numerical value
If n: even, $x_{md} = \frac{(n/2)\text{value} + (n/2+1)\text{value}}{2}$

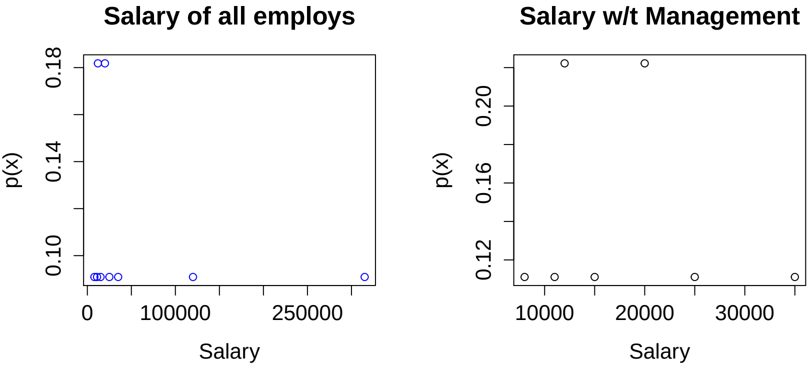
- Data Set (n observations,numerical values): $x_1, x_2, \ldots, x_k$ corresponding frequencies $f_1, f_2, \ldots, f_k$

Sample mode

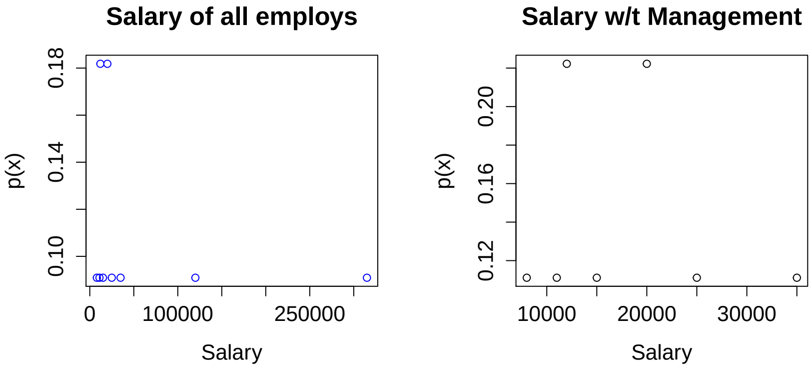
$x_{mode} = x_j$ with f_j is the greatest frequency

- Note: No single value occurs most frequently, i.e. the greatest frequency = $f_j = f_l = \ldots = f_p$, then $x_{mode} = x_j = x_l = \ldots = x_p$.

Mean, Median, and Mode



Mean, Median, and Mode



A rough Guideline

Measurement Scale	Best Measure of the "Middle"
Nominal (Categorical)	Mode
Symmetrical data	Mean
Skewed data	Median

Sample Statistics, Cont.

- Data Set (n observations,numerical values): $x_1, x_2, \ldots, x_n$

Sample variance

$$\sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} \quad \text{or}$$

$$\sigma^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n\,\bar{x}^2\right)$$

- Standard deviation: +ve square root of sample variance

SD

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

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- Standard deviations: Rs. 92198.11

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- Standard deviations (w/t Employs 9,10): Rs. 8472.18

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- Standard deviations: Rs. 92198.11
- Standard deviations (w/t Employs 9,10): Rs. 8472.18

Sample Statistics, Cont.

- Data set: $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$
- Question: Any statistics to determine a relationship between x_i and y_i?
- Answer: Sample correlation coefficient (r_{xy})

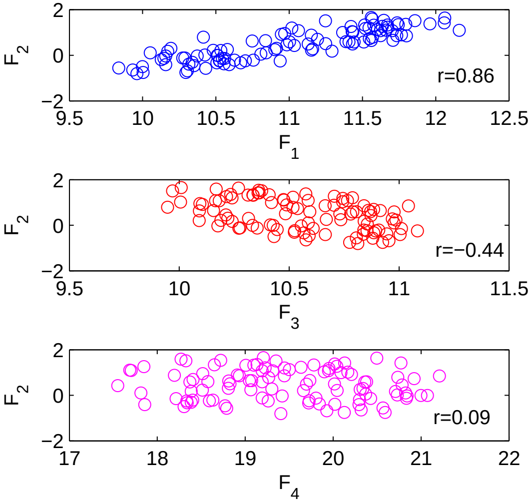
Sample variance

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{x})}{(n-1)\sigma_x \sigma_y}$$

- $r_{xy} > 0 \rightarrow$ +ve correlation, $r_{xy} > 0 \rightarrow$ -ve correlation
- $|r_{xy}$= measure of the strength of the linear relationship between x and y variables

Sample Statistics, Cont.

- Four flow variables: F_1, F_2, F_3, F_4



Percentiles

- Sample $100p$, $0 \leq p \leq 1$ percentile:
  100p percent of the data values are less or equal to it

Example

Course grades: 10% E, 16% D, 24% C, 32% B, 15% A, 5% S
I got B grade, what is percentile for A?

- Computation of percentile:
  - For Group data: **Add up all percentages below the particular group, plus half the percentage at the group**
  - For a data set of size n, $100p$ percentile:
    - At least np of the values are less than equal to it
    - If two data values satisfy this condition, then the arithmetic average of these two values

Quartiles

- Values that divide a list of numbers into quarters:
- Computation of Quantiles:
  - Put the list of numbers in order
  - Then cut the list into four equal parts
  - The Quartiles are at the "cuts"

Example

6, 7, 4, 5, 4, 2, 8
Order them: 2,4,4,5,6,7,8
Quartile 1 (Q1) = 4
Quartile 2 (Q2) = 5
Quartile 3 (Q3) = 7

Indian Institute of Technology Madras

Module II

Nirav Bhatt
Email: niravbhatt@iitm.ac.in

Statistical Data Analysis

Module I

- Descriptive statistics: Data analysis through
  - Numerical computation of sample statistics: Mean, variance, mode, range, ...
  - Graphical representation: Organize, summarize, and visualize in terms of different types of graphs, Box plots, scattered plots...

Statistical Data Analysis

Module II

- Module II- Inference or inductive statistics: Data analysis for decision making
  - Parameter estimation: Determine unknown parameters from sample data
  - Hypothesis testing: Verify or validate a postulate (or hypothesis) regarding population(s) or parameters using the data

Module II

Statistical Hypothesis testing and confidence intervals

- Topics:
  - Point estimation of parameters
  - Confidence interval computation
  - Statistical hypothesis testing
- Learning Outcomes: Students should be able to
  - estimate parameters from observations
  - compute confidence intervals
  - formulate statistical hypothesis and run tests using data

Data types, form and variables

- Format: Images, texts, numbers, videos...
- Types:
  - Numerical (or quantitative)
    - Interval: Ordering of scale and difference between two values in data is meaningful
      GATE Scores, IQ Scores, credit score
    - Ratio: Interval with clear definition of absolute zero
      Height in meters, Weight in Kg, Concentrations...
  - Categorical (or qualitative)
    - Nominal: Categories with no order
      Patient's name or ID, color of t-shirt...
    - Ordinal: Categories with order but no different between values
      Grades, Weight in Healthy, overweight, obese

Parameter Estimation

- Example: Lethal dose of a medicine
- Important to know for assessing the overall efficacy of the medicine
- Variability due to Gender, BMI, Age, Geography...
- FDA needs a representative value or a range for lethal dose of a medicine
- Use sample data to compute a reasonable value of lethal dose
  Point estimate of lethal dose

Hypothesis testing

- Example: Two medicines A and B for a disease
- Scientist conjectures that A is better medicine for the disease
- How can you prove or reject the conjecture?
- If the scientists can perform experiments on different sets of patients having the same disease with both medicines and shows that A is better
- Need to collect data and a procedure to show that A is better medicine than B
  Statistical hypothesis testing
- Emphasis on the better medicine

Parameter Estimation

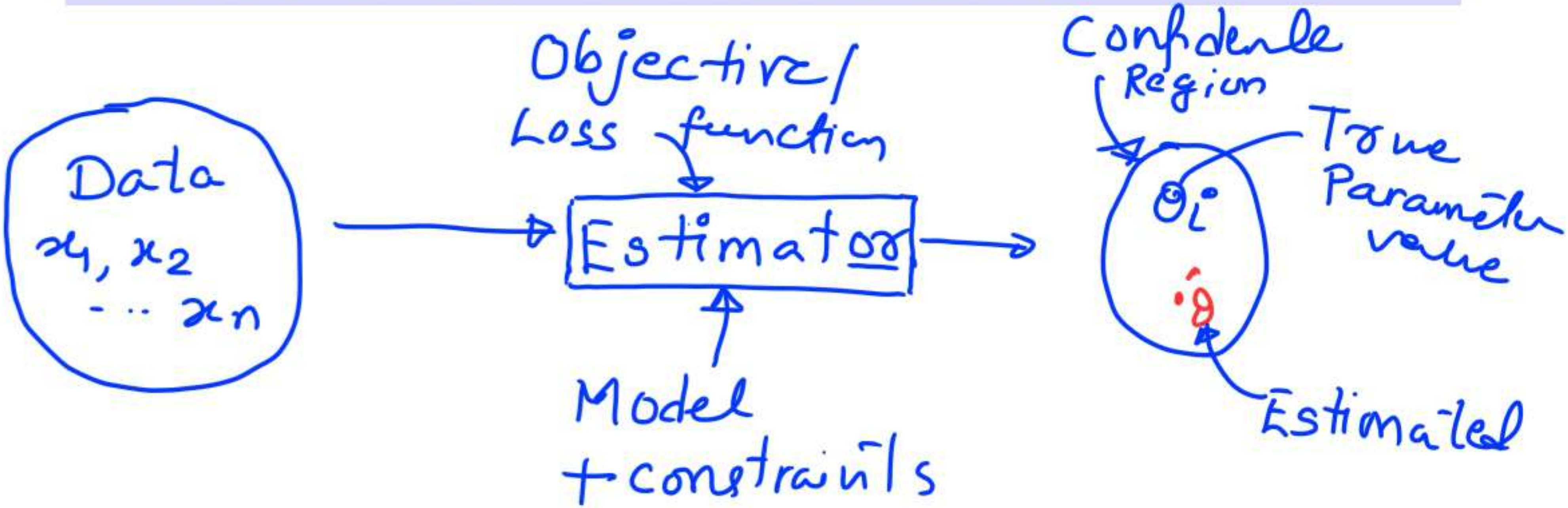
- Two problems of parameter estimation
  - Estimate parameters of a distribution from data — μ, σ^2
  - Estimate parameters of models from data $y = \alpha_1 x_1 + \alpha_2 x_2$, Estimate α_1, α_2
- Objectives of estimation: (i) Estimating parameters, and (ii) provide a goodness of estimated parameters
- Parameter estimation involves two steps:
  - Estimating parameters using methods for estimation
  - Assessing the "goodness" of the estimated parameters and provide bounds on variables

Parameter Estimation

Elements

Definition : Estimator

It is the process of inferring unknown parameters in a model or distribution from a given set of data and other information using a *mathematical map* between the unknowns parameters and the known information and a decision criterion.



Parameter Estimation

Types

- Point estimators: Produce single-valued estimates (more common)
  Examples: kinetic parameter estimates from data, mean height of person in the classroom, expected life of a mobile device....
- Interval estimators: Produce an interval
  Examples: catalyst particle size, age of the students in BT5450
- Other types: Non-parametric, Parametric, and semi-parametric
  Depends on the information available such as function and/or density distribution forms

Parameter Estimation

Random Sample

Random Sample

Consider RVs $X_1, X_2, \ldots, X_n$. These RVs are random sample of size n if
(i) the X_i's are independent RVs
(ii) item Each X_i is drawn from the same probability distribution

Random sample should be
- Representation of population
- Bias free

Parameter Estimation

Statistics

Statistics

A statistic is any function of the observation in a random sample, $\hat{\Theta} = g(X_1, X_2, \ldots, X_n)$

- $\hat{\Theta}$ is a random variable
- Example:

$$\underbrace{\begin{array}{l} m \\ \text{Random samples} \end{array}}\left\{\begin{array}{l} \{x_1^1, x_2^1, \ldots, x_n^1\} \longrightarrow \hat{\mu}_1 \\ \{x_1^2, x_2^2, \ldots, x_n^2\} \longrightarrow \hat{\mu}_2 \\ \quad \vdots \qquad\qquad\qquad\qquad\quad \vdots \\ \{x_1^m, x_2^m, \ldots, x_n^m\} \longrightarrow \hat{\mu}_m \end{array}\right\} \text{Different values}$$

means

Parameter Estimation

Sampling distribution

Sampling distribution

The probability density function of a statistic is called a sampling distribution

→ sample mean, $\bar{x}$ computed from random sample: $x_1, \ldots, x_n$ for a population with $N(\mu, \sigma^2)$

→ Sampling distribution of $\bar{x}$

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Parameter Estimation

Point Estimator

- Random sample: $X_1\ X_2, \ldots, X_n$ with $f(x, \boldsymbol{\theta})$: Density function
- $\boldsymbol{\theta}$: Unknown parameters in column- vector form

Point Estimator

A point estimate of some population parameters $\boldsymbol{\theta}$ is a single numerical vector-value $\hat{\boldsymbol{\theta}}$ of a statistic. The statistic is called the point estimator.

Normal distribution:

$$f(x,\theta) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{with } \theta = \begin{bmatrix} \mu \\ \sigma^2 \end{bmatrix}$$

objective fn

Data $\longrightarrow$ $\boxed{\text{Estimator}}$ $\longrightarrow$ $\hat{\mu} = 2$

$f(x,\theta)$ $\longrightarrow$ $\qquad\qquad\qquad\quad$ $\hat{\sigma}^2 = 3.1$

Parameter Estimation

Estimator

- Statistical properties of the estimate
  - Accuracy: How accurate is the estimate on the average?
  - Precision: Variability of the estimates obtained from different random samples?
- The given estimator gives an estimate with the least variability?
- What about true value of θ (θ_t) and $\hat{\theta}$ obtained from the estimator?
- How does the sample size n affect the value of estimate?

Role of n:

$n \to \infty$ (Large sample size)

$\hat{\theta} \to \theta_t$?

Parameter Estimation

Unbiased Estimators

- How accurate is the estimate on the average?
  Closeness of estimate to the true values
- How close values can be computed using an Estimator $\hat{\Theta}$?

$$E(\hat{\Theta}) = \boldsymbol{\theta}_t$$

Unbiased estimator

A point estimator $\hat{\Theta}$ is an unbiased estimator for the parameter $\boldsymbol{\theta}$ if

$$E(\hat{\Theta}) = \boldsymbol{\theta}_t$$

Parameter Estimation

Unbiased Estimators

Bias of an estimator

If the estimator is not unbiased estimator, the bias (b) can be computed as

$$b = E(\hat{\Theta}) - \theta_t$$

$E(\hat{\theta}) = \theta_b$, Then

Bias, $b = \theta_b - \theta_t$

If $\boxed{\theta_b \cong \theta_t}$, $\boxed{\hat{\theta} \rightarrow \text{unbiased Estimator}}$

$b = 0$ &

Parameter Estimation

Example: Show that sample mean and variance are unbiased

Random samples : $X_1, X_2, \ldots, X_n$

$X_i \sim P(\mu, \sigma^2)$ P: Distribution

$i = 1, \ldots, n$

sample mean, $\bar{X} = \sum_{i=1}^{n} \frac{X_i}{n}$

$$E[\bar{X}] = E\left[\sum_{i=1}^{n} \frac{X_i}{n}\right] = \frac{1}{n} E\left[\sum_{i=1}^{n} X_i\right]$$

$$= \frac{1}{n} E[X_1 + X_2 + \ldots + X_n]$$

$$= \frac{1}{n} \{E[X_1] + E[X_2] + \ldots + E[X_n]\}$$

$$= \frac{1}{n} [\mu + \mu + \ldots + \mu] = \mu$$

$$\text{Bias} = E[\bar{X}] - \mu = \mu - \mu = 0$$

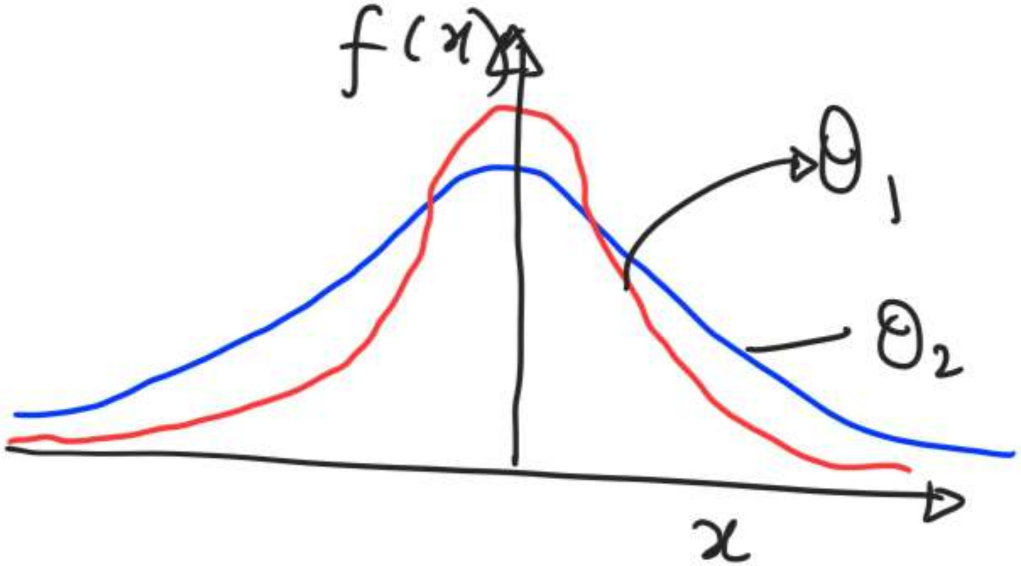
Parameter Estimation

Variance of a Point Estimator

- Two unbiased estimators $\hat{\Theta}_1$ and $\hat{\Theta}_2$

$$E(\hat{\Theta})_1 = \boldsymbol{\theta}_t, \quad E(\hat{\Theta})_2 = \boldsymbol{\theta}_t$$

Question: Which one to choose?



$Var(\Theta_1) < Var(\Theta_2)$

choose the estimator with minimum variance

Parameter Estimation

Variance of a Point Estimator

Minimum variance unbiased estimator (MVUE)

Consider all the unbiased estimators (say total m) of θ $(\hat{\Theta}_1, \hat{\Theta}_2, \ldots, \hat{\Theta}_{m,})$, the one with the smallest variance is called MVUE.

$$\mathrm{Var}(\hat{\Theta}_2) < \mathrm{Var}(\hat{\Theta}_1) < \ldots \mathrm{Var}(\hat{\Theta}_m)$$

Θ_2 is MVUE

Parameter Estimation

Standard Error

- Precision and variability of an estimate?
  Standard error of the estimate

Standard Error of an Estimator

The standard error of of an estimator $\hat{\Theta}$ is its standard deviation given by

$$\hat{\sigma}_{\hat{\Theta}} = \sqrt{\text{Var}(\hat{\Theta})}$$

$E[\hat{\theta}] = \mu$, $var(\hat{\theta}) = \sigma^2$

$\hat{\sigma}_{\hat{\theta}} = \sigma$

$\rightarrow$ Point estimate $\pm$ standard error

$\mu \pm \sigma$

Parameter Estimation

Mean Squared error of an Estimator

- Only biased estimators are available
  How to select an estimator?
- Means squared error of an estimator $\hat{\Theta}$ of the parameter θ

Means squared error

$$\text{MSE}(\hat{\Theta}) = E[(\hat{\Theta} - \theta_t)^2]$$

or

$$\begin{aligned}\text{MSE}(\hat{\Theta}) &= (E[\hat{\Theta} - E(\hat{\Theta})])^2 + (\theta_t - E(\hat{\Theta}))^2 \\ &= Var(\hat{\Theta}) + (\text{Bias})^2\end{aligned}$$

- Bias-variance trade-off : MSE($\hat{\Theta}$) is total of variance & Bias

Parameter Estimation

Mean Squared error of an Estimator

- Two estimators of the parameter θ: MSE($\hat{\Theta}_1$) and MSE($\hat{\Theta}_2$)
- Relative efficiency of estimators

$$\frac{\text{MSE}(\hat{\Theta}_1)}{\text{MSE}(\hat{\Theta}_2)}$$

- Relative efficiency < 1: $\hat{\Theta}_1$ is a more efficient $\hat{\Theta}_2$

Parameter Estimation

Methods of Point Estimation

- Method of moments:
  Equate population moments to sample moments

Random Sample: $X_1, X_2, \ldots, X_n$ from a PMF or PDF with unknown p parameters $\boldsymbol{\theta}$. The moment estimators $\hat{\Theta}_1, \ldots, \hat{\Theta}_p$ can be found by equating the first p population moments to the first p sample moments and solving the set of nonlinear equations

Parameter Estimation

Methods of Point Estimation

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k^{th} moment: $E[X^k]$; $k=1$; $E[X] \to$ mean

$$\underbrace{E[X^k]}_{k^{th}\text{ moment for population}} = \underbrace{\hat{\theta}^k(x_1, x_2, \ldots x_n)}_{k^{th}\text{ moment computed from data}}$$

Parameter Estimation

Method of moments : Example

Data: $x_1, x_2 \dots x_n$; Drawn from $Exp(\lambda)$

λ: Unknown parameter. Estimate λ using MoM

$$E[x^1] = \frac{1}{n}\sum_{i=1} x_i$$

$$\frac{1}{\lambda} = \frac{1}{n}\sum x_i$$

$$\Rightarrow \lambda = \frac{n}{\sum x_i}$$

Parameter Estimation

Maximum Likelihood Estimation

- RV $X \sim f(x, \theta)$, θ: Unknown parameters
- Observations $x_1, x_2, \ldots, x_n$
- The likelihood function of the sample is

$$L(\theta) = L(\theta / x_1, x_2, \ldots, x_n) = f(x_1, \theta) \cdot f(x_2, \theta) \cdot f(x_3, \theta) \cdot \ldots \cdot f(x_n, \theta)$$

$\max\limits_{\theta} L(\theta)$: Maximum likelihood estimator

$$\max_{\theta} L(\theta) = \max_{\theta} \prod_{i=1}^{n} f(x_i, \theta)$$

Parameter Estimation

Maximum Likelihood Estimation

Data: $x_1, \ldots x_n \sim$ Bernoulli R. V.

$$\text{PMF}: f(x, \theta) = p^{x}(1-p)^{1-x}, \quad x = 0,1$$
$$= 0 \quad , \text{otherwise}$$

Parameter to be estimated: p

$$\theta = [p] = p$$

$\rightarrow$ Construct $L(\theta)$

$$L(\theta) = f(x_1, p) \cdot f(x_2, p) \cdot \ldots f(x_n, p)$$
$$= p^{x_1}(1-p)^{1-x_1} \cdot p^{x_2}(1-p)^{1-x_2} \cdot \ldots \cdot p^{x_n}(1-p)^{1-x_n}$$
$$= p^{\sum x_i}(1-p)^{n-\sum x_i}$$

Parameter Estimation

Maximum Likelihood Estimation

$$L(p) = p^{\sum x_i} (1-p)^{n - \sum x_i}$$

$$\Leftrightarrow \ln L(p) = \sum x_i \ln p + (n - \sum x_i) \ln(1-p)$$

$$\frac{\partial L}{\partial p} = \frac{\partial \ln L(p)}{\partial p} = 0$$

$$\frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p} = 0 \Rightarrow p = \frac{\sum x_i}{n}$$

Verify $\dfrac{\partial^2 \ln L(p)}{\partial p} < 0$

Parameter Estimation

Maximum Likelihood Estimator: Properties

- Unbiased estimator : For Large n
- Variance of $\hat{\Theta}$ is nearly as small as the one that could be obtained with any other estimator
- $\hat{\Theta}$: An approximate normal distribution

Invariance Property

$$\left.\begin{array}{c} \theta_1, \theta_2 \cdots \theta_p \\ \downarrow \text{MLE} \\ \hat{\theta}_1, \hat{\theta}_2, \cdots \hat{\theta}_p \end{array}\right\} \longrightarrow \begin{array}{c} h(\theta_1, \cdots \theta_p) \\ \text{MLE} \\ h(\hat{\theta}_1, \hat{\theta}_2 \cdots \hat{\theta}_p) \end{array}$$

Parameter Estimation

Bootstrapping Estimation

- Non-parametric approach of estimation *Unlike MLE and MoM:*

  No need for assumption about underlying distribution

- Often used for computing standard error and confidence intervals for relatively small sample size

- Uses sampling with replacement strategies

R.S. = $\{1, 2, 8, 9, 3, 6, 7\}$

sample 1: 1, 8, 9, 3, 9

Sample 2: 8, 8, 3, 6, 6

Parameter Estimation

Bootstrapping Estimation

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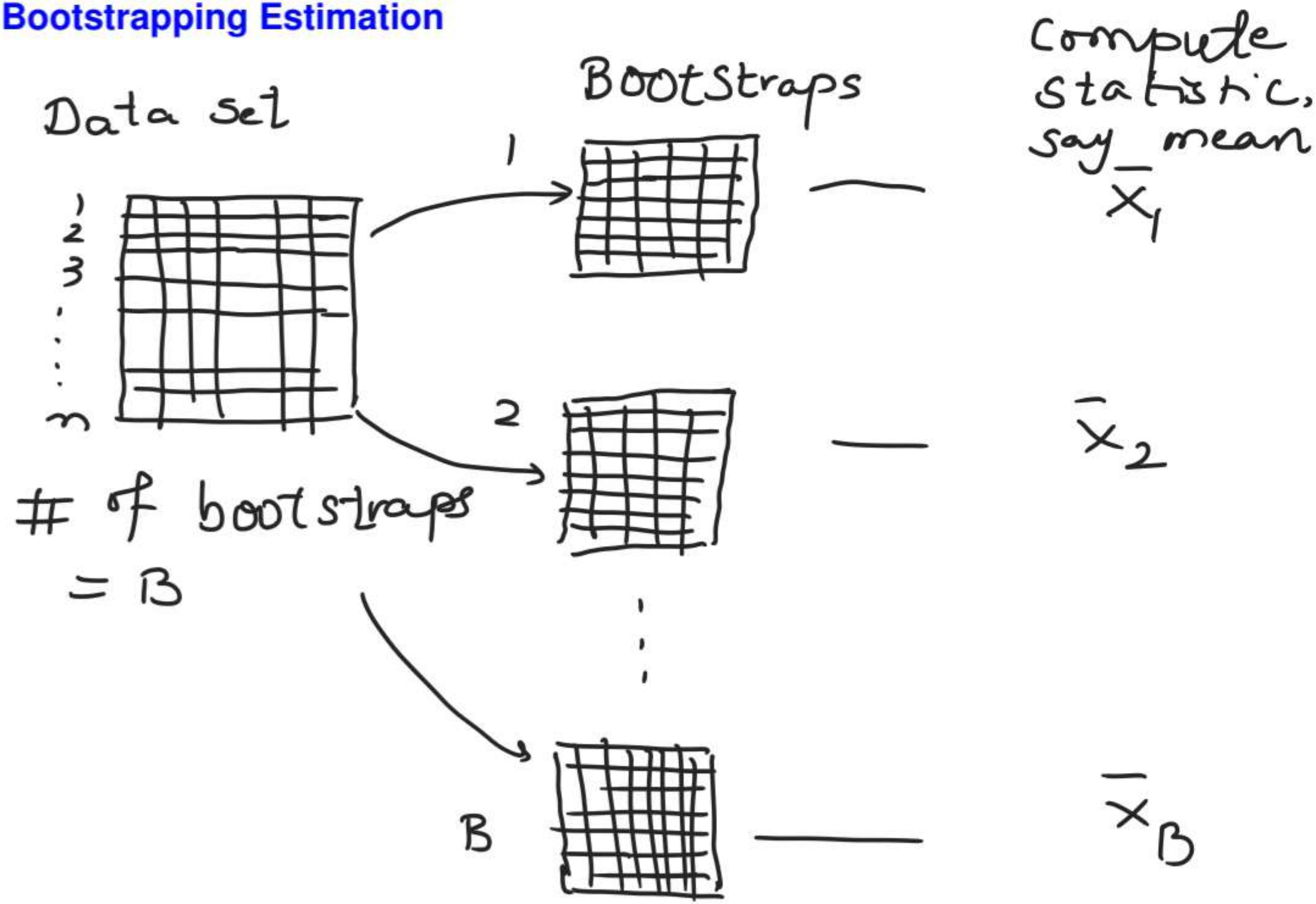
- Samples: $X_1, X_2, \ldots, X_n$ drawn from independent and identical but unknown distribution
- Let $\hat{\Theta} = \hat{\Theta}(X_1, X_2, \ldots, X_n)$ be statistic

Parameter Estimation

Bootstrapping Estimation



Parameter Estimation

Bootstrapping Estimation

- Bootstrap means

$$
\begin{aligned}
\bar{X}_1 &= \quad \text{mean}(X_1^{*,1}, \ldots, X_n^{*,1}) \\
\bar{X}_2 &= \quad \text{mean}(X_1^{*,2}, \ldots, X_n^{*,2}) \\
&\vdots \\
\bar{X}_B &= \quad \text{mean}(X_1^{*,B}, \ldots, X_n^{*,B})
\end{aligned} \tag{1}
$$

- Bootstrap estimate of the variance

$$
\text{var}(\bar{X}) = \frac{1}{B-1} \sum_{i=1}^{B} (\bar{X}_i - \bar{X}_B)^2, \text{ with } \bar{X}_B = \frac{1}{B} \sum_{i=1}^{B} \bar{X}_i
$$

Confidence interval

Introduction

- Point estimate: How close to true value?
- Interested in knowing the variability of the population parameters
- Range of plausible values: Confidence interval
- An interval estimate for a population parameter is called a confidence interval
- *Confidence*: Specifies level of confidence 90%,95%, 99%
- Constructed so that it contains true unknown population parameter(s)

Confidence Interval
Introduction

- Random sample: $X_1, X_2, \ldots, X_n$
- Unknown Distribution, Unknown mean μ and Known variance σ^2
- Sample mean $\bar{X} \sim F(\mu, \sigma^2/n)$
- Standardize $\bar{X}$, New R.V.

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Three types of intervals
1. Confidence Interval
2. Tolerence Interval
3. Prediction Interval

Confidence Interval

Introduction

- Confidence interval: lower and upper bounds,

$$l \leq \mu \leq u \qquad l, u: \text{Unknown}$$

- l and u: End points computed from the data
- l and u: Values of random variable L and U
- Question: How do we determine values of l and u

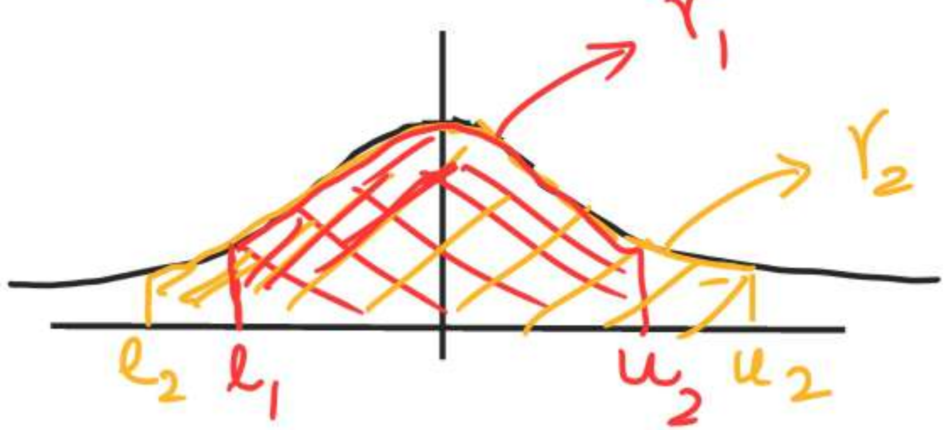
Random sample: $x_1, x_2 \ldots, x_n$

$$l = L(x_1, x_2, \ldots x_n)$$

$$u = U(x_1, x_2, \ldots x_n)$$

Confidence Interval

Introduction



- L and U: RVs
- Determine values of these RVs such that

$$P\{L \leq \mu \leq U\} = \gamma = 1 - \alpha, \quad 0 \leq \gamma, \alpha \leq 1$$

- From samples $x_1, x_2, \ldots, x_n$, l and u can be computed to determine CI with (1-α) probability

$$l \leq \mu \leq u$$

l and u: Lower and upper-confidence bounds

Confidence Interval

Introduction

Normal distribution $X \sim N(\mu, \sigma^2)$

Data: $X_1, X_2, \ldots, X_n$

μ is unknown and σ^2: known

Objective: Find interval for unknown μ.

Standard Normal R.V., $Z = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

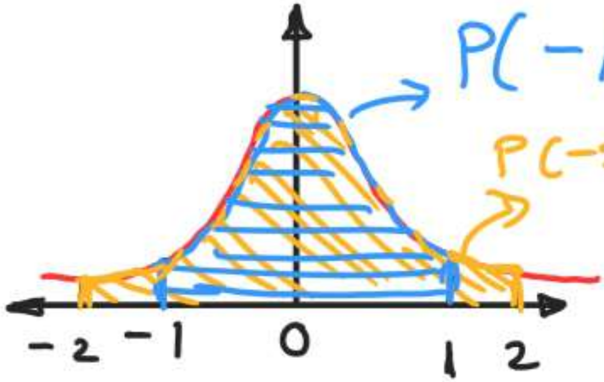
compute $\bar{x}$ from data

$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n}$$

Confidence Interval

Introduction

z- distribution :



$P(-1 \le z \le 1) = 0.68$
$= 1 - 0.32$

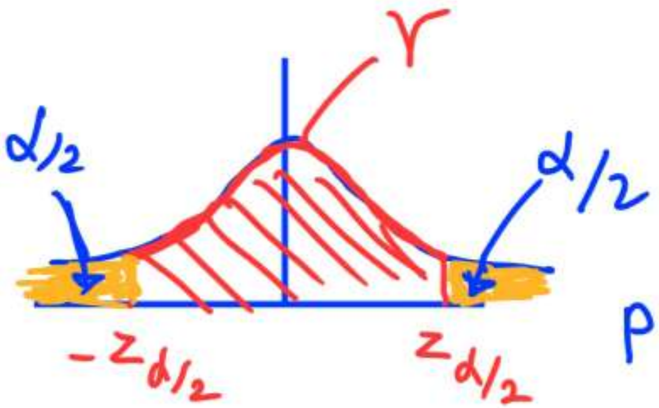
$P(-2 \le z \le 2) = 0.96$
$= 1 - 0.04$

For $z \in [-1, 1]$, $\alpha = 0.32$
$z \in [-2, 2]$, $\alpha \cong 0.04$

Given α, $z_{\alpha/2}$ can be computed from z-tables

$$P\left(-z_{\alpha/2} \le Z \le z_{\alpha/2}\right) = \gamma = 1 - \alpha$$

$$P\left(-z_{\alpha/2} \le \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right) = 1 - \alpha$$



$$P\left(\underbrace{\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}_{l} \le \mu \le \underbrace{\bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}_{u}\right) = 1 - \alpha$$

Confidence Interval

CI and Precision

- 90% or 95% or 99%?
- $z_{\alpha/2}$ for $\alpha = 0.05$ and $\alpha = 0.01$

$$z_{0.025} = 1.96$$

$$z_{0.005} = 2.58$$

- Length of confidence intervals

$$95\%, \text{ Length} = 2(1.96\sigma/\sqrt{n}) = 3.92\sigma/\sqrt{n}$$

$$99\%, \text{ Length} = 2(2.58\sigma/\sqrt{n}) = 5.16\sigma/\sqrt{n}$$

Confidence Interval

Introduction

One-sided CI on the mean, Known σ^2

$100(1-\alpha)\%$ upper-confidence bound

$$\mu \le \overline{X} + z_\alpha \sigma/\sqrt{n}$$

Lower-confidence bound

$$\mu \ge \overline{X} - z_\alpha \frac{\sigma}{\sqrt{n}}$$

Confidence Interval

Introduction

Large Sample, n, CI for μ

- σ^2 unknown but $n \sim$ large
- compute sample variance, s^2

$$Z \cong \bar{X} - \mu / s/\sqrt{n} \quad \text{(Approximately z-distribution)}$$

- CI

$$\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}} \le \mu \le \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}}$$

- Typically, in practice, $n \geq 40$

Confidence Interval

Introduction

- Error=$\|\bar{x} - \mu\|$
- For given σ, specify E, and α, then n: number of samples required

$$n = \left(\frac{z_{\alpha/2}\sigma}{E} \right)^2$$

- For example, E=0.5, $\sigma = 2$, $\alpha = 0.05$, Then n

$$n = \left(\frac{(1.96)(2)}{0.5} \right)^2 = 61.5$$

- For $E = 0.25$, $n =?$ $\sigma = 2$, $\alpha = 0.05$, $n \cong 246$
- $\sigma = 1$, $n =?$ $E = 0.5$, $\alpha = 0.05$, $n = 16$
- For $\alpha = 0.01$, $n =?$, $z_{0.005} = 2.58$, $n = 107$

Confidence Interval

One-sided Confidence Bounds

- One-sided Confidence Bounds for a given α: Provides
  - Lower bound $l \leq \mu$
  - Upper bound $u \geq \mu$
- For a given *alpha* Computed by
  - Lower bound $\bar{x} - z_\alpha \sigma/\sqrt{n} \leq \mu \leq \infty$
  - Upper bound $\bar{x} + z_\alpha \sigma/\sqrt{n} \geq \mu \geq -\infty$

Confidence Interval

Unknown Population variance

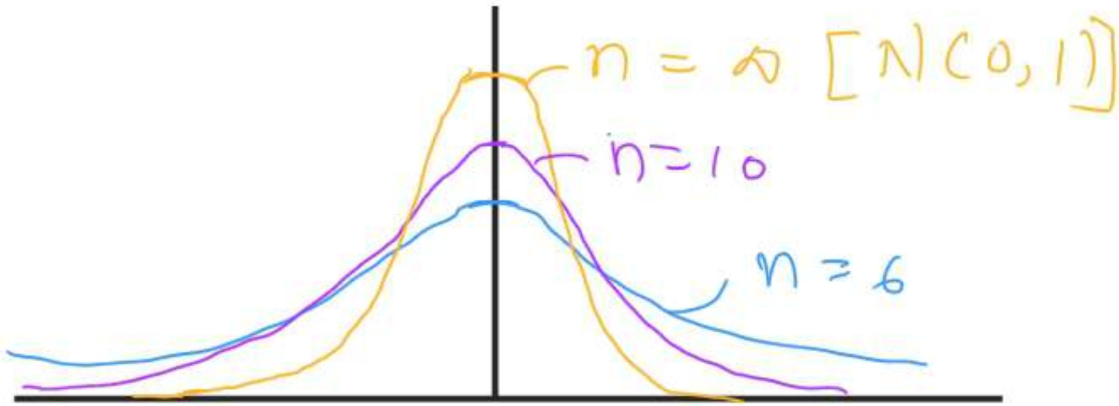
- So far
  - n random samples, unknown μ, and known σ^2
- n random samples, unknown μ, and σ^2?
- Confidence interval for μ
- Sample variance, S^2 can be computed from n observations
- A statistic can be computed (on same line as z-statistic, $Z = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}}$)

$$T = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}}$$

- T is RV from t-distribution with $n-1$ degrees of freedom

$$f(u) = \frac{\Gamma[n/2]}{\sqrt{\pi(n-1)}\,\Gamma(n-1/2)} \frac{1}{[(\frac{u^2}{n-1})+1]^{n/2}} \quad -\infty < u < \infty$$

t - distribution

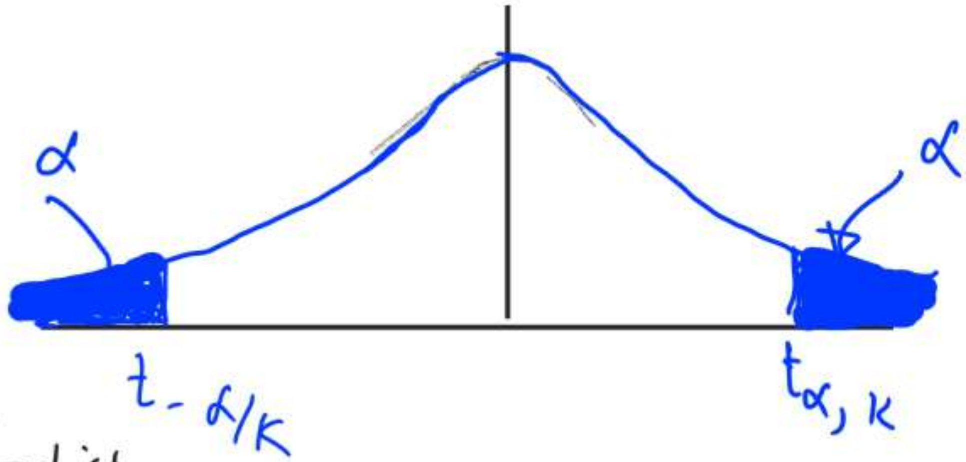


Confidence Interval

Unknown Population variance

- $T \sim$ t-dist. with $n-1$ degrees of freedom
- symmetric like z-dist.
- CI for μ with unknown σ^2
- $100(1-\alpha)$% CI

$$\bar{x} - t_{\frac{\alpha}{2}, n-1} s/\sqrt{n} \le \mu \le \bar{x} + t_{-\alpha/2, n-1} s/\sqrt{n}$$



- Large n, z-dist. $\sim$ t-dist.

Confidence Interval

CI for σ^2 of a Normal Distribution

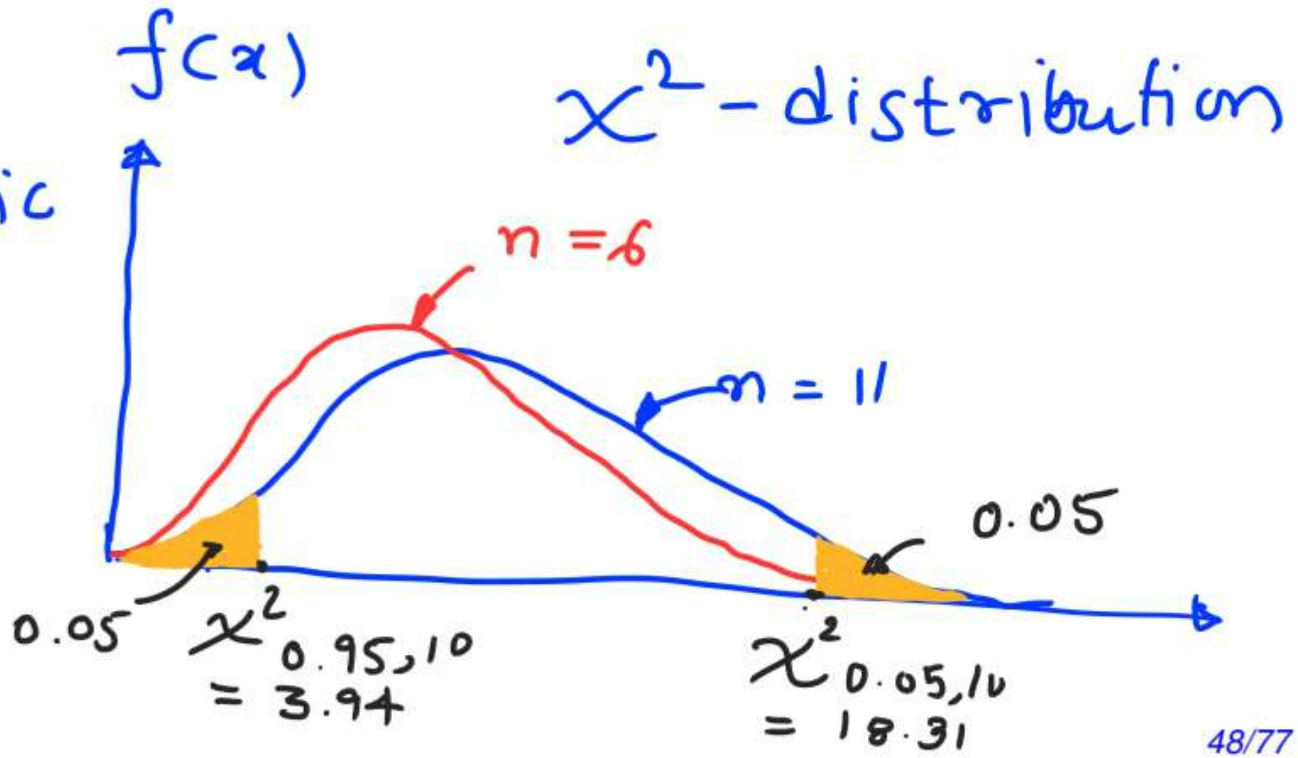
- χ^2-distribution for n samples from $\mathcal{N}(\mu, \sigma^2)$
- X^2-statistic (or RV)

$$X^2 = \frac{(n-1)S^2}{\sigma^2}$$

- $X^2 \sim \chi^2_{n-1}$

- Not symmetric

$$P(X^2 > \chi^2_{\alpha,k}) = \int_{\chi^2_{\alpha,k}}^{\infty} f(u)\,du = \alpha$$



Confidence Interval

confidence Interval on the variance

Two-sided $100(1-\alpha)\%$ with n observations

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}, \quad s^2: \text{ computed sample variance}$$

Interpretation of CI:

- L & V : Random Variables
- CI : Random interval
- Interpretation: If large number of random samples are collected then $100(1-\alpha)\%$ of these CI will contain the true value of statistic (mean, variance)

Confidence Interval

Parameter of interest	symbol	Other parameter	Confidence Interval $100(1-\alpha)\%$
Mean: Normal distribution	μ	σ^2: known	$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
Mean: Arbi. distribution large size	μ	σ^2: Not known compute s^2 from data	$\bar{x} - z_{\alpha/2}\frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2}\frac{s}{\sqrt{n}}$
Mean: Normal distribution	μ	σ^2: Not known compute s^2 from data	$\bar{x} - t_{\alpha/2, n-1}\frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha/2, n-1}\frac{s}{\sqrt{n}}$
Variance: Normal distribution	σ^2	Mean μ unknow and estimate μ & s^2	$\frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2, n-1}}$

Confidence Interval

Summary : $100(1-\alpha)\%$

Parameter of interest	Lower-bound	Upper-bound
μ, both σ^2 known & σ^2 unknown but large n	$\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}} \leq \mu$	$\bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}} \geq \mu$
μ & σ^2 unknown	$\bar{x} - t_{\alpha,n-1} \frac{s}{\sqrt{n}} \leq \mu$	$\bar{x} + t_{\alpha,n-1} \frac{s}{\sqrt{n}} \geq \mu$
σ^2 & μ unknown	$\frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} \leq \sigma^2$	$\sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha,n-1}}$

Hypothesis Testing

Example

- Treatments for a disease: T-A and T-B
- Claim: T-A is better than T-B
- Practitioners question: Does T-A better than T-B?
- Approach to answer practitioners question: **Hypothesis testing: Decision making process**

Hypothesis Testing
Introduction

- Claim: T-A is better than T-B
- Claim(s) or statement(s): Statistical Hypothesis (es)
  Statement about the parameters of one or more populations
- Claim: T-A is better than T-B: Claim to population's parameter
- Practitioners' interest: mean number of days to recuperate from the appearance of clinical symptoms

Hypothesis Testing

Introduction

- Practitioners' interest: mean number of days to recuperate from the appearance of clinical symptoms

  Population mean

- μ_{T-A} and μ_{T-B}: mean days to recuperate for treatment T-A and T-B
- $\mu_{T-A} > \mu_{T-B}$
- Formal re-casting of statement as two hypotheses:

$$H_0 : \mu_{T-A} = \mu_{T-B}$$

$$H_1 : \mu_{T-A} > \mu_{T-B}$$

- H_0: Null hypothesis: Both treatments are same
- H_1: Alternative hypothesis: T-A is better than T-B

Hypothesis Testing

Introduction

- One-sided alternative hypothesis

$$H_0: \mu_{T-A} = \mu_{T-B} \quad H_1: \mu_{T-A} > \mu_{T-B}$$

or

$$H_0: \mu_{T-A} = \mu_{T-B} \quad H_1: \mu_{T-A} < \mu_{T-B}$$

- Claim: Mean number of days to recuperate for T-A is 8 days or $\mu_{T-A} = 8$
- Two-sided alternative hypothesis

$$H_0: \mu_{T-A} = 8 \ \ days \quad H_1: \mu_{T-A} \neq 8 \ \ days$$

- By convention: Null hypothesis is an equality claim

Hypothesis Testing
Elements

- *Hypothesis: A statement about the population or model or distribution not about the sample*

  Use sample to verify hypothesis

- Truth or falsity of a claim (or hypothesis) is never known in practical situation

- Hypothesis testing: Probabilistic approach to reach a conclusion based on population parameter(s)

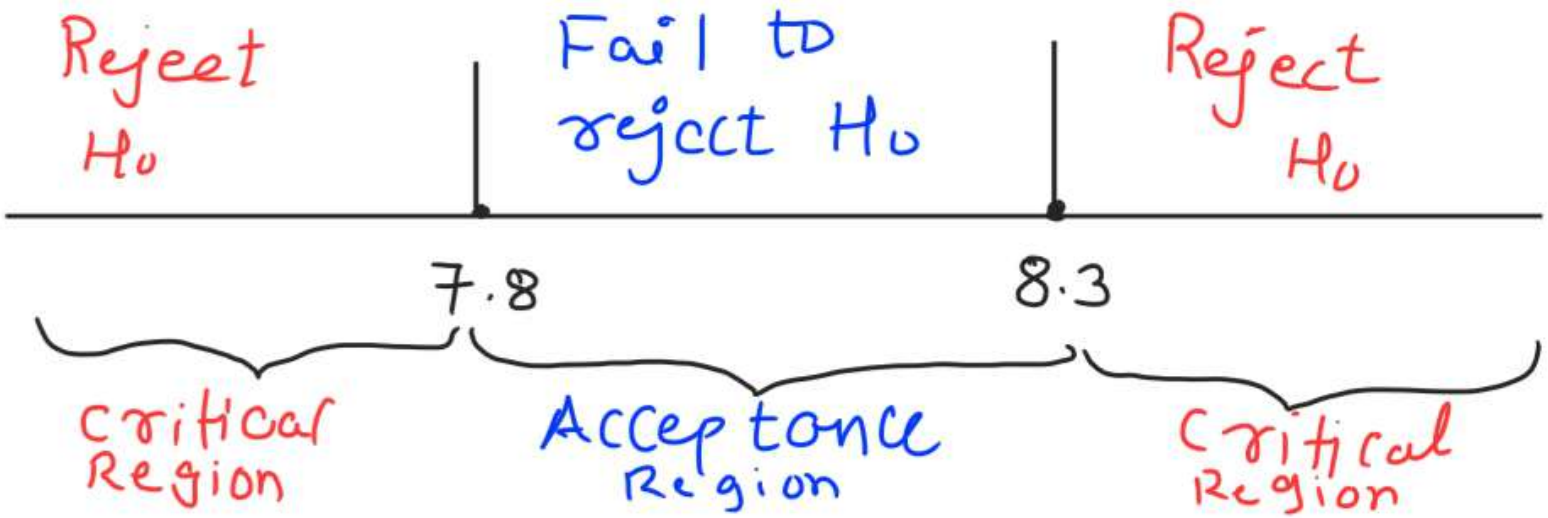
Hypothesis Testing

Elements

- Two-sided alternative hypothesis

$$H_0 : \mu_{T-A} = 8 \;\; days \quad H_1 : \mu_{T-A} \neq 8 \;\; days$$

- Samples are available for T-A different patients
- Sample mean: Take on many different values
- Let us define range (recall CI): $7.8 \leq \bar{x} \leq 8.3$
- Acceptance region: any value in the range
- Critical regions: outside the acceptance region



Hypothesis Testing

Elements

- Pitfall I:
  - $\mu_{T-A} = 8$: Truth
  - Random sample selected: $\bar{x} = 8.7$
  - Outside acceptance region ($7.8 \leq \bar{x} \leq 8.3$)
  - Reject H_0 in favor of H_1
    Wrong conclusion $\rightarrow$ Type I error

Type I Error

Rejecting H_0 when it is true is defined as a type I error

Type II Error

Failing to reject H_0 when it is false is defined as a type II error

Hypothesis Testing

Elements

Decision	H_0 is true	H_0 is false
Fail to reject H_0	No error	Type II error
Reject H_0	Type I error	No error

- Quantifying Type I and II errors
- Type I error: Probability of rejecting H_0 when H_0 is true

Hypothesis Testing

Elements

Decision	H_0 is true	H_0 is false
Fail to reject H_0	No error	Type II error
Reject H_0	Type I error	No error

- Quantifying Type I and II errors
- Type I error: Probability of rejecting H_0 when H_0 is true

Hypothesis Testing

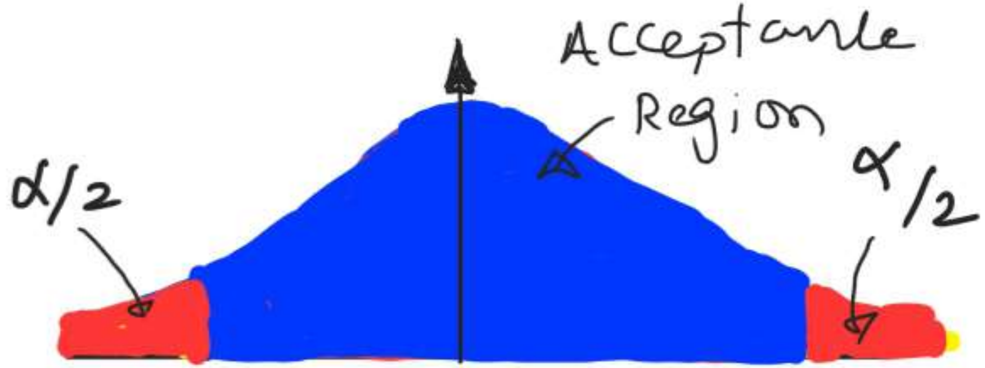
Elements

Type I error

Rejecting the H_0 when it is true is defined as a type I error.

Probability of Type I error

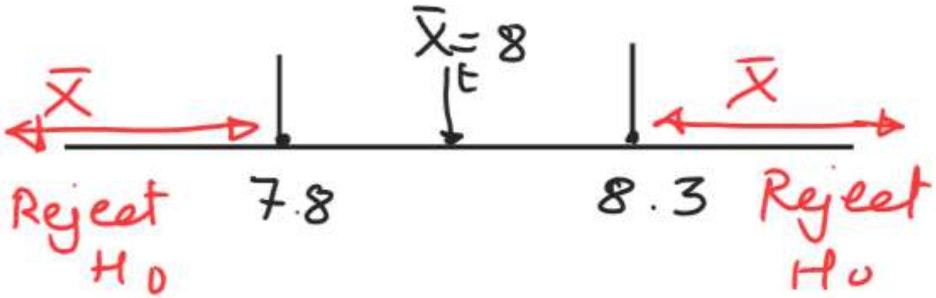
α= P(type I error)=P(reject H_0 when H_0 is true)



Hypothesis Testing

Elements

$$P(\text{Type I error}) = P(\text{Reject } H_0 \text{ when } H_0 \text{ is true})$$



$$\sigma = 1, \quad n = 100$$

$$P(\text{Type I error}) = P(\bar{X} \le 7.8 \text{ when } \bar{X}_t = 8) + P(\bar{X} \ge 8.3 \text{ when } \bar{X}_t = 8)$$

$$= P\left(Z_1 \le \frac{7.8 - 8}{1/10}\right) + P\left(Z_2 \ge \frac{8.3 - 8}{1/10}\right)$$

$$= \alpha$$

Hypothesis Testing

Elements

Type II error

β=P(Probability of Type II error)=P(fail to reject H_0 when H_0 is false)

claim: Average weight of students: 50 kg

$H_0: \mu = 50$, $H_1: \mu \neq 50$

True mean, $\mu = 52$

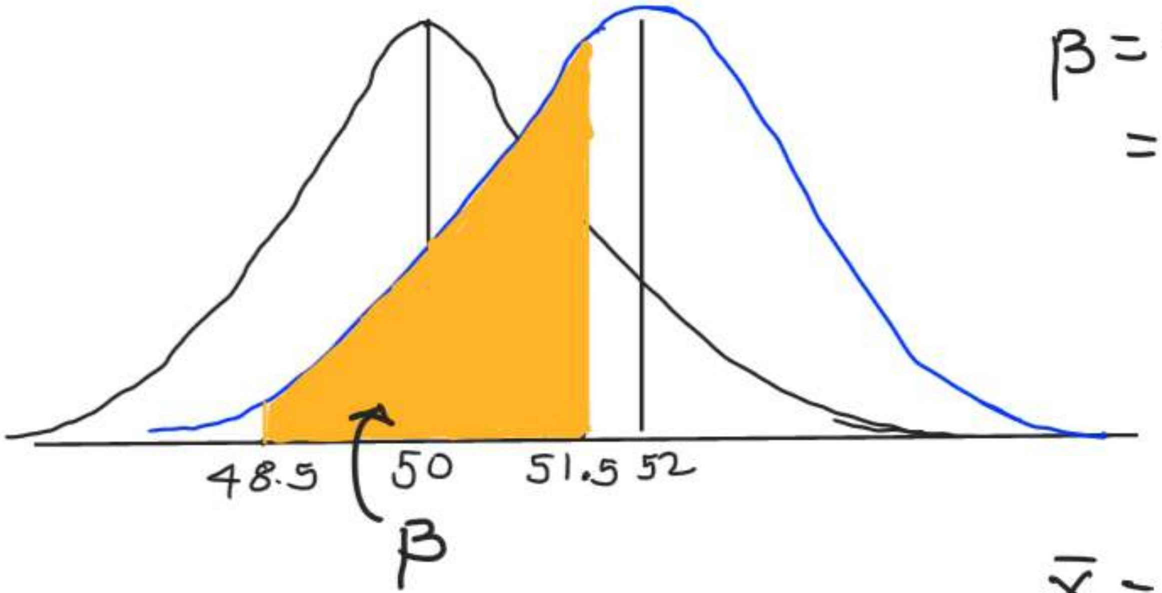
Acceptance region: $48.5 \leq \bar{X} \leq 51.5$

$\beta = P(48.5 \leq \bar{x} \leq 51.5$ when $\mu = 52)$

Hypothesis Testing

Elements

$H_0: \mu = 50 \quad H_1: \mu = 52$



$$\beta = P(48.5 \le X \le 50.5, \mu = 52)$$
$$= P(X \le 51.5) - P(X \le 48.5)$$

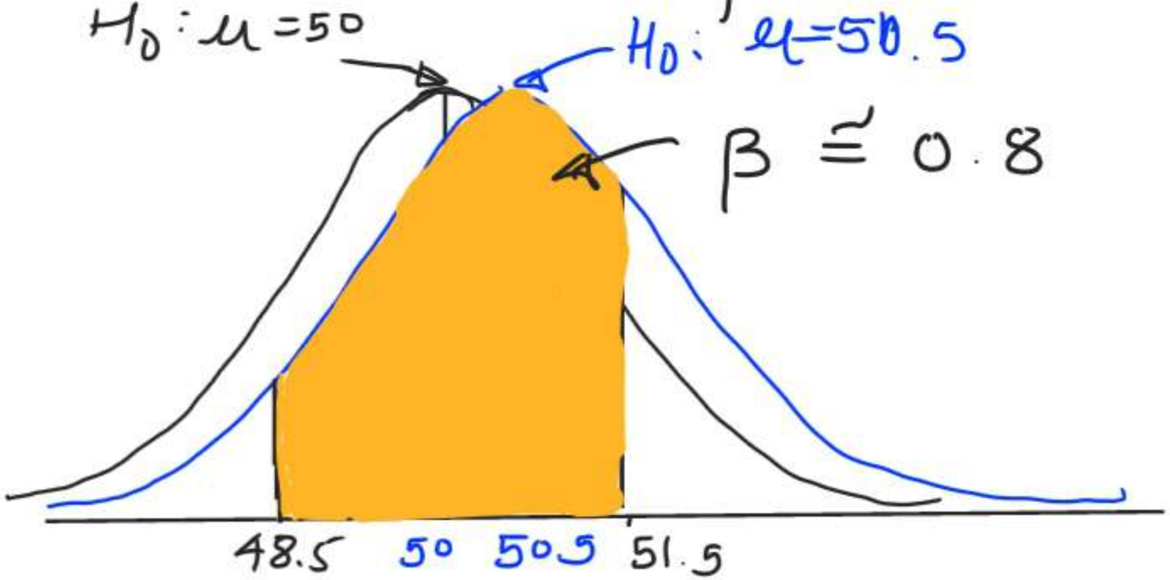
For $\sigma/\sqrt{n} = 0.8$, $z_1 = \dfrac{\bar{x} - 52}{\sigma/\sqrt{n}}$

$$\beta = P(z_1 \le -0.62) - P(Z_1 \le -2.5)$$
$$= 0.26$$

Hypothesis Testing

Elements

Instead of $\mu = 52$, $\mu = 50.5$



$$\beta = P(48.5 \le x \le 51.5,\ \mu = 50.5)$$
$$\cong 0.8$$

Type II error is higher when $\mu = 50.5$.

Hypothesis Testing

Elements

Important points

1. The size of the critical region can be reduced by type I error, α.

2. For given sample size, n, decrease in the probability of one type error results in an increase in the other type.

3. For given α, increase n reduces β

4. value of β decreases as the diff. between the true mean and the hypothesized value increases

Hypothesis Testing

Elements

- β: Not constant, depends on true value of parameter and sample size
- Extent of falsity of null hypothesis
- Accept H_0: Weak conclusion
- Failing to reject H_0: Strong conclusion

↳ Does not mean
"H_0 is true", but, it means
"more data are required to make strong conclusion".

Hypothesis Testing

Power

- Power:
  Power of statistical test: Probability of rejecting null Hypothesis H_0 when H_1 is true
- Power $=1-\beta$
- Power: Probability of correctly rejecting a false H_0

→ Power is used to compare two statistical test

→ Useful measure of sensitivity of a statistical test.

Hypothesis Testing

Elements

one-sided hypothesis:

claim involving phrases "greater than", less than or at least ...

"one-sided hypothesis testing"

— Appropriate alternative hypothesis test has to be chosen.

— one-sided H_1; Rejecting H_0 is a strong conclusion.

Hypothesis Testing
Elements

- α: Fixed significance level
- α: Doesn't provide any idea location of parameters in critical region

P-value

The P-value is the smallest level of significance that would lead to rejection of H_0 with the given data.

P-value: Observed significance level

— Provides how significant the data are

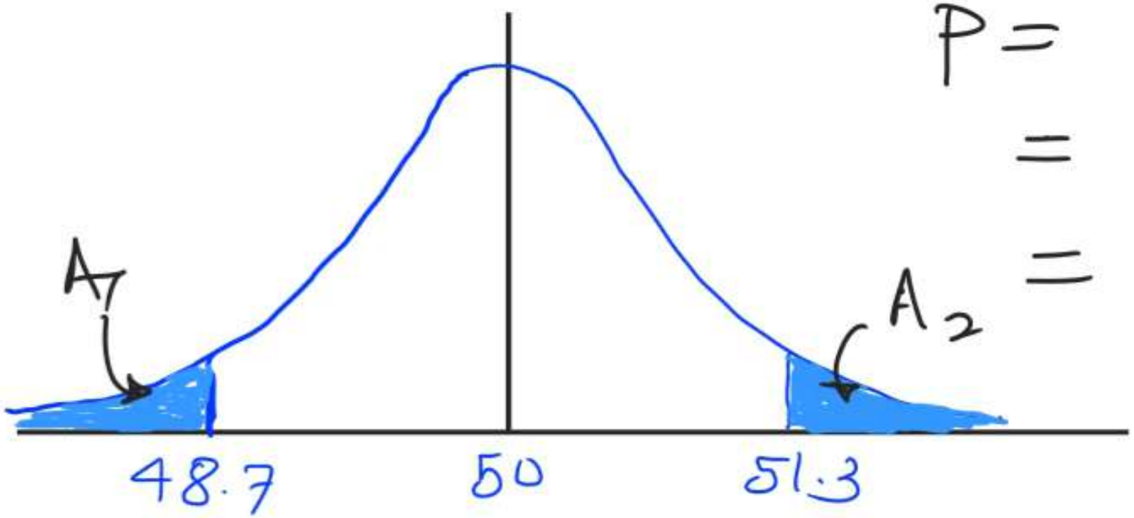
Hypothesis Testing

Elements

Two-sided hypothesis test

$H_0: \mu = 50$, $H_1: \mu \neq 50$, $n = 16$, $\sigma = 2.5$

- Observed sample mean, $\bar{x} = 51.3$



$$P = A_1 + A_2$$
$$= 1 - P(48.7 < \bar{X} < 51.3)$$
$$= 1 - 0.962 = 0.038$$

It indicates $\bar{x} = 51.3$ is a rare event when $p = 0.038$

Hypothesis Testing

General Procedure for Hypothesis tests

1. Parameter of interest: Identify the parameter of interest for a context
2. Null hypothesis, H_0: State the null hypothesis
3. Alternative hypothesis, H_1: Specify an appropriate alternative hypothesis
4. Test statistic: Determine an appropriate test statistic
5. Reject H_0 if: State the rejection criteria for the null hypothesis
6. Computations: Compute any necessary sample quantities and value of test statistic
7. Draw conclusions: Decide whether or not H_0 should be rejected and report that in the problem context